

Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-27

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Action Center

Run date: 2026-05-27 **Universe:** 660 ETFs / 26 clusters

Full PC1: LONG since 2026-04-13 (D+30, 5.0%) **Clusters long:** 11/26 **ETFs long:** 319/660 (48%)

Today's events (cluster level)

$\text{spread_buy} = 5\text{dma} - 50\text{dma}$; $\text{spread_sell} = 5\text{dma} - 20\text{dma}$ (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

No new cluster crossovers today.

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **$\text{spread_buy} = 5\text{dma} - 50\text{dma}$, $\text{spread_sell} = 5\text{dma} - 20\text{dma}$** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 4 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_24	Cash / Ultra-Short Fixed Income	0.321	1	1.00	1.000	0.99
cluster_22	Japan Equity	7.579	1	1.00	1.000	0.97
cluster_14	Financials	2.552	1	1.00	0.428	0.22
cluster_4	U.S. Small Cap Equity	3.168	1	0.99	0.192	0.11

Full Sample PC1

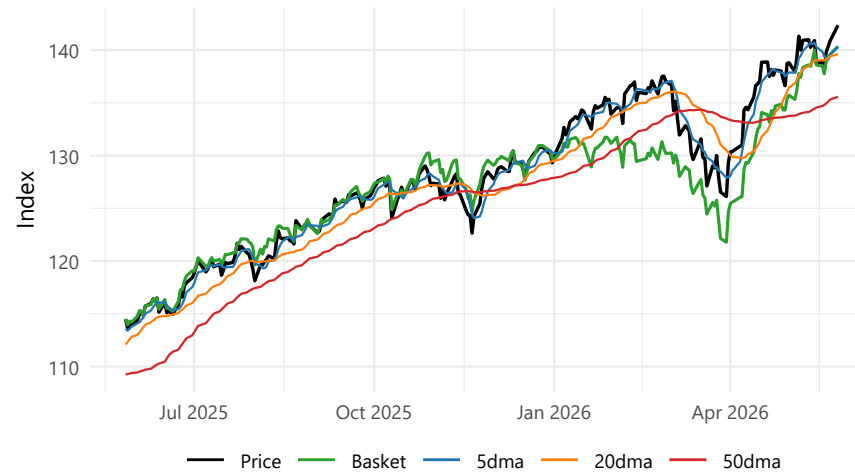


Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: LONG since 2026-04-13 (30 days, 5.0%)

5dma: 140.266 20dma: 139.621 50dma: 135.566

Spread vs 50dma: +4.701 Spread vs 20dma: +0.645

P(buy crossover within 5 days): 100.0% P(sell): 49.1%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
V00	cluster_1	U.S. Large Cap Equity	0.944	30	13.1%	7.71%	0.98
VHT	cluster_11	Healthcare	0.915	13	0.3%	7.68%	0.93
VEA	cluster_3	Developed ex-US Equity	0.883	31	3.5%	5.57%	0.99
IGV	cluster_16	Technology	0.854	28	8.0%	5.12%	0.91
SRLN	cluster_23	Bank Loan / Floating Rate	0.848	34	1.3%	6.21%	0.89
FLOT	cluster_23	Bank Loan / Floating Rate	0.848	34	1.3%	5.86%	0.89
BKLN	cluster_23	Bank Loan / Floating Rate	0.848	34	1.3%	4.18%	0.89
FLRN	cluster_23	Bank Loan / Floating Rate	0.848	34	1.3%	2.55%	0.89
FLTR	cluster_23	Bank Loan / Floating Rate	0.848	34	1.3%	0.70%	0.89
VTV	cluster_2	U.S. Large Cap Value	0.792	30	3.5%	11.20%	0.98
VNQ	cluster_15	Real Estate	0.720	31	5.0%	8.25%	0.99
IJH	cluster_6	U.S. Mid Cap Equity	0.493	31	9.7%	5.59%	0.96
IEMG	cluster_8	Emerging Markets Equity	0.259	31	10.3%	4.39%	0.99
PDBC	cluster_20	Broad Commodities	0.237	94	41.0%	4.15%	0.98
XLP	cluster_19	Consumer Staples	0.138	1	-0.8%	5.98%	0.99
XLU	cluster_19	Consumer Staples	0.138	1	-0.8%	3.72%	0.99

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_14	Financials	1.00	0.216	1.097	0.96
EWJ	cluster_22	Japan Equity	1.00	0.973	0.908	0.96
UUP	cluster_26	Currency	1.00	1.000	0.517	0.83
ICSH	cluster_24	Cash / Ultra-Short Fixed In...	1.00	0.989	0.418	0.84
SHV	cluster_24	Cash / Ultra-Short Fixed In...	1.00	0.989	0.347	0.84
SQQQ	cluster_26	Currency	1.00	1.000	0.339	0.83
MINT	cluster_24	Cash / Ultra-Short Fixed In...	1.00	0.989	0.175	0.84
GBIL	cluster_24	Cash / Ultra-Short Fixed In...	1.00	0.989	0.118	0.84
IJR	cluster_4	U.S. Small Cap Equity	0.99	0.109	0.978	0.99

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
26	cluster_26	Currency	Currency	5	FLAT				1.000	-0.000
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				0.989	+0.080
22	cluster_22	Japan Equity	Japan Equity	9	FLAT				0.973	+0.050
1	cluster_1	U.S. Large Cap Equity	U.S. Large Cap Eq...	98	LONG		30	13.1%	0.944	+0.149
11	cluster_11	Healthcare	Healthcare	17	LONG	shifted	13	0.3%	0.915	+1.487
3	cluster_3	Developed ex-US Equity	Developed ex-US E...	61	LONG		31	3.5%	0.883	+0.247
16	cluster_16	Technology	Technology	14	LONG	shifted	28	8.0%	0.854	+0.533
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	LONG		34	1.3%	0.848	+0.045
2	cluster_2	U.S. Large Cap Value	U.S. Large Cap Va...	88	LONG	shifted	30	3.5%	0.792	+0.802
15	cluster_15	Real Estate	Real Estate	14	LONG	shifted	31	5.0%	0.720	+0.647
6	cluster_6	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	38	LONG	shifted	31	9.7%	0.493	+0.097
8	cluster_8	Emerging Markets Equity	Emerging Markets ...	25	LONG	shifted	31	10.3%	0.259	+1.072
20	cluster_20	Broad Commodities	Broad Commodities	10	LONG		94	41.0%	0.237	-0.423
14	cluster_14	Financials	Financials	14	FLAT	shifted			0.216	+1.216
19	cluster_19	Consumer Staples	Consumer Staples	11	LONG		1	-0.8%	0.138	-0.040
4	cluster_4	U.S. Small Cap Equity	U.S. Small Cap Eq...	52	FLAT	shifted			0.109	-0.281
9	cluster_9	Materials	Materials	21	FLAT	shifted			-0.157	-0.220
10	cluster_10	High Yield Bond	High Yield Bond	21	FLAT	shifted			-0.377	-0.075
12	cluster_12	China Equity	China Equity	15	FLAT	shifted			-0.639	-1.243
7	cluster_7	Global Equity	Global Equity	30	FLAT	shifted			-0.718	-1.073
25	cluster_25	International Fixed Income	International Fix...	6	FLAT				-0.766	+0.036
18	cluster_18	Gold	Gold	12	FLAT				-0.799	+0.043
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.919	+0.023
13	cluster_13	Fixed Income	Fixed Income	15	FLAT	shifted			-0.981	-0.302
17	cluster_17	Municipal Bond	Municipal Bond	13	FLAT	shifted			-0.986	+0.014
5	cluster_5	Aggregate Fixed Income	Aggregate Fixed I...	43	FLAT				-0.990	+0.010

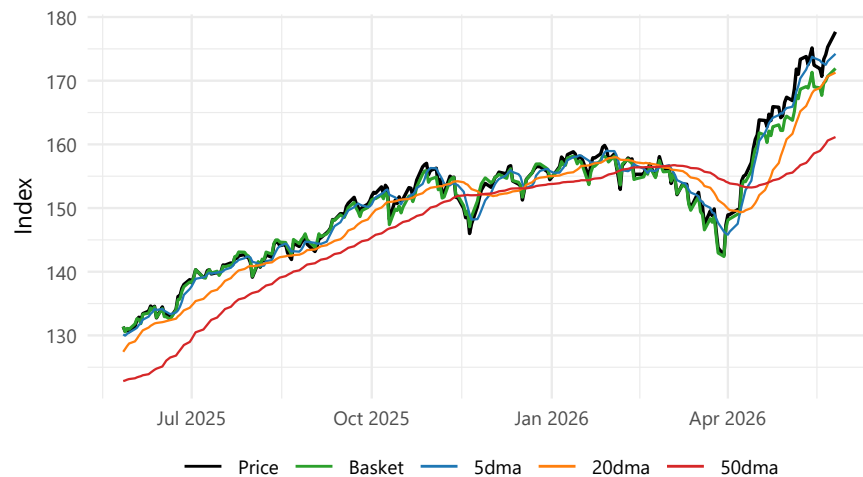
Cluster Details

Cluster 1 – U.S. Large Cap Equity

Members: 98 **Var explained:** 0.877 **Signal:** LONG since 2026-04-13 (30 days, 13.1%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +32.2%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +37.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
Technology	22	0.2198
U.S. Large Cap Growth	11	0.1172
U.S. Large Cap Equity	9	0.0970
U.S. Total Market Equity	7	0.0756
Global Equity	7	0.0741
Consumer Discretionary	6	0.0569
U.S. Equity	4	0.0425
Asset Allocation	4	0.0389

U.S. Mega Cap Equity	3	0.0321
U.S. Quality / Factor Equity	3	0.0311
Communication Services	3	0.0284
U.S. ESG Equity	2	0.0215
U.S. Multifactor Equity	2	0.0212
Growth Equity	2	0.0210
Large Cap Blend Equities	1	0.0107
Commodity	1	0.0107
U.S. Mega Cap Growth	1	0.0106
Large Cap Value Equities	1	0.0106
U.S. Equal Weight Equity	1	0.0105
Options Income	1	0.0097
U.S. Small Cap Technology	1	0.0094
Alternatives	1	0.0093
Trend-Following Equity	1	0.0091
International IPO Equity	1	0.0085
U.S. Trend-Following Equity	1	0.0082
Israel Equity	1	0.0080
Singapore Equity	1	0.0074

Replication basket – VOO (cor=0.98) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
VOO	1.106	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	SPY	0.0108	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
68	IWV	0.0108	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
69	IYY	0.0108	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
72	SPYX	0.0108	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
74	ESGU	0.0108	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
75	IWB	0.0108	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
76	IWL	0.0108	iShares Russell Top 200 ETF	U.S. Equity	LONG
80	VTHR	0.0108	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
83	VONE	0.0108	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
84	MGC	0.0108	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
85	VV	0.0108	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
86	ILCB	0.0108	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
87	ITOT	0.0108	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG

88	IVV	0.0108	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
89	SCHB	0.0108	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
90	SCHX	0.0108	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
91	SPTM	0.0108	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
92	VOO	0.0108	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
93	VTI	0.0108	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
11	QQQ	0.0107	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
14	ACWI	0.0107	iShares MSCI ACWI ETF	Global Equity	LONG
19	URTH	0.0107	iShares MSCI World ETF	Global Equity	LONG
42	ONEQ	0.0107	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
44	IWF	0.0107	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
45	IVW	0.0107	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
48	VOOG	0.0107	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
53	VONG	0.0107	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
55	ILCG	0.0107	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
56	IUSG	0.0107	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
59	VUG	0.0107	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
66	DSI	0.0107	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
67	SUSA	0.0107	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
71	OEF	0.0107	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
81	GSLC	0.0107	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
20	CRBN	0.0106	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
21	SPGM	0.0106	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG
22	VT	0.0106	Vanguard Total World Stock Index ETF	Global Equity	LONG
41	NULG	0.0106	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
43	IWY	0.0106	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
54	MGK	0.0106	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
57	SCHG	0.0106	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
58	SPYG	0.0106	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
60	TILT	0.0106	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	LONG
65	JHML	0.0106	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
70	LRGF	0.0106	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
73	XLG	0.0106	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	LONG
77	QUAL	0.0106	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
82	EPS	0.0106	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
15	AOA	0.0105	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
25	QQEW	0.0105	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
37	IGM	0.0105	iShares Expanded Tech Sector ETF	Technology	LONG
40	IYW	0.0105	iShares U.S. Technology ETF	Technology	LONG
47	VGT	0.0105	Vanguard Information Tech ETF	Technology	LONG
52	FTEC	0.0105	Fidelity MSCI Information Technology In...	Technology	LONG
63	IOO	0.0105	iShares Global 100 ETF	Global Equity	LONG
1	XLK	0.0104	State Street Technology Select Sector S...	Technology	LONG
29	XT	0.0104	iShares Future Exponential Technologies...	Technology	LONG
36	PWB	0.0104	Invesco Large Cap Growth ETF	Growth Equity	LONG

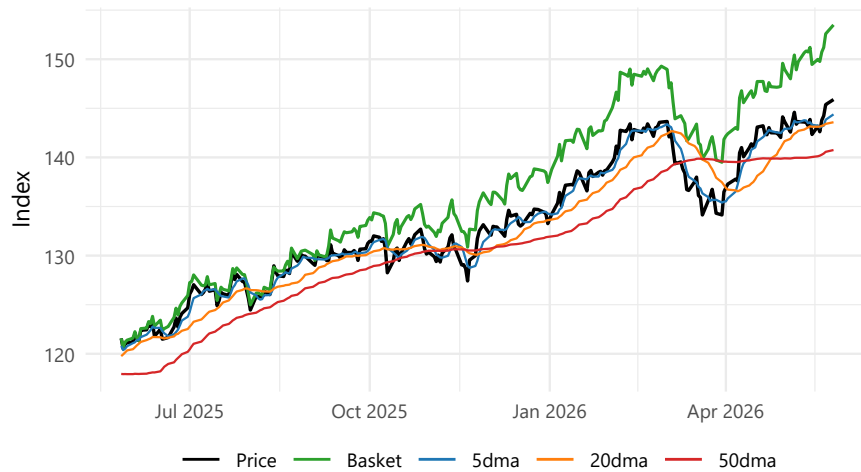
38	IXN	0.0104	iShares Global Tech ETF	Technology	LONG
64	CATH	0.0104	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
8	AIQ	0.0103	Global X Artificial Intelligence & Tech...	Technology	LONG
31	FXL	0.0103	First Trust Technology AlphaDEX	Technology	LONG
32	XNTK	0.0103	State Street SPDR NYSE Technology ETF	Technology	LONG
18	AOR	0.0102	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
26	QTEC	0.0102	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
61	SPHB	0.0102	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
78	QUS	0.0102	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	LONG
79	SPHQ	0.0102	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
28	TDIV	0.0101	First Trust NASDAQ Technology Dividend ...	Technology	LONG
7	ROBO	0.0100	ROBO Global Robotics and Automation Ind...	Technology	LONG
39	IYC	0.0100	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
46	VCR	0.0099	Vanguard Consumer Discretionary ETF	Consumer Discretionary	FLAT
51	FDIS	0.0099	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	FLAT
2	XLY	0.0098	State Street Consumer Discretionary Sel...	Consumer Discretionary	FLAT
6	BOTZ	0.0098	Global X Robotics & Artificial Intellig...	Technology	LONG
23	NXTG	0.0098	First Trust Indxx NextG ETF	Technology	LONG
35	QYLD	0.0097	Global X NASDAQ 100 Covered Call ETF	Options Income	FLAT
98	QTUM	0.0097	Defiance Quantum ETF	Technology	LONG
5	SMH	0.0096	VanEck Semiconductor ETF	Technology	LONG
49	VOX	0.0096	Vanguard Communication Services ETF	Communication Services	LONG
50	FCOM	0.0096	Fidelity MSCI Communication Services In...	Communication Services	LONG
4	SOXX	0.0095	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
17	AOM	0.0095	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
30	XSD	0.0094	State Street SPDR S&P Semiconductor ETF	Technology	LONG
95	PSCT	0.0094	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
3	XLC	0.0093	State Street Communication Services Sel...	Communication Services	FLAT
12	QAI	0.0093	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
27	PSI	0.0093	Invesco Semiconductors ETF	Technology	LONG
97	FTXL	0.0093	First Trust Nasdaq Semiconductor ETF	Technology	LONG
34	PTNQ	0.0091	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
24	PRNT	0.0089	3D Printing (The) ETF	Technology	LONG
96	RTH	0.0089	VanEck Retail ETF	Consumer Discretionary	FLAT
16	AOK	0.0087	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
9	ESPO	0.0085	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
33	FPXI	0.0085	First Trust International Equity Opport...	International IPO Equity	LONG
62	PTLC	0.0082	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG
94	EIS	0.0080	iShares MSCI Israel ETF	Israel Equity	FLAT
13	EWS	0.0074	iShares MSCI Singapore ETF	Singapore Equity	LONG

Cluster 2 – U.S. Large Cap Value

Members: 88 **Var explained:** 0.874 **Signal:** LONG since 2026-04-13 (30 days, 3.5%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +23.2%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +21.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)

**Category breakdown**

category	n_members	abs_weight
U.S. Dividend Equity	24	0.2716
U.S. Large Cap Value	10	0.1197
Industrials	6	0.0672
U.S. Multifactor Equity	4	0.0461
Value Equity	4	0.0461
Dividend Equity	4	0.0446
U.S. Low Volatility Equity	4	0.0427
U.S. Mid Cap Equity	3	0.0348
U.S. Value Factor Equity	3	0.0347
Utilities	3	0.0321
U.S. Fundamental Equity	2	0.0239

U.S. Mid Cap Value	2	0.0238
U.S. Equal Weight Equity	2	0.0237
U.S. Equity	2	0.0237
U.S. Small Cap Low Volatility	2	0.0217
U.S. Mega Cap Value	1	0.0118
U.S. Revenue Weighted Equity	1	0.0118
Large Cap Blend Equities	1	0.0118
U.S. Size Factor Equity	1	0.0117
U.S. Buyback Equity	1	0.0115
U.S. Large Cap Equity	1	0.0115
U.S. Quality / Factor Equity	1	0.0112
U.S. Mid Cap Low Volatility	1	0.0112
U.S. Moat Equity	1	0.0110
U.S. Defensive Equity	1	0.0108
Global Equity	1	0.0105
Communication Services	1	0.0097
Real Estate	1	0.0092

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.005

ticker	rep_wt	name	category	etf_signal
VTV	1.024	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
34	IWD	0.0121	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
51	SCHV	0.0121	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
4	RSP	0.0120	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
18	PRF	0.0120	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
37	JPUS	0.0120	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
44	VONV	0.0120	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
50	IUSV	0.0120	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	LONG
54	VTV	0.0120	Vanguard Value ETF	U.S. Large Cap Value	LONG
23	CFA	0.0119	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
26	NULV	0.0119	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	LONG
30	FNDX	0.0119	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
33	ONEV	0.0119	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
36	IVE	0.0119	iShares S&P 500 Value ETF	U.S. Large Cap Value	LONG
41	VOOV	0.0119	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
47	VYM	0.0119	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	LONG

48	ILCV	0.0119	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	LONG
52	SPYV	0.0119	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	LONG
66	IWS	0.0119	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
70	VOE	0.0119	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
77	DTD	0.0119	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
82	DGRO	0.0119	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
86	JPME	0.0119	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
8	FEX	0.0118	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
19	RWL	0.0118	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
21	AIVL	0.0118	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
32	IWX	0.0118	iShares Russell Top 200 Value ETF	Value Equity	LONG
45	MGV	0.0118	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
55	FTA	0.0118	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
72	PFM	0.0118	Invesco Dividend Achievers ETF	U.S. Dividend Equity	LONG
76	DLN	0.0118	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
31	EQAL	0.0117	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
38	SIZE	0.0117	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
67	ONEY	0.0117	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	LONG
25	NOBL	0.0116	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	LONG
27	SDY	0.0116	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
28	FDVV	0.0116	Fidelity High Dividend ETF	U.S. Dividend Equity	LONG
35	IWR	0.0116	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
49	IMCB	0.0116	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
83	VIG	0.0116	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
87	JVAL	0.0116	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
5	FVD	0.0115	First Trust VL Dividend	U.S. Dividend Equity	LONG
7	PKW	0.0115	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
16	IYJ	0.0115	iShares U.S. Industrials ETF	Industrials	FLAT
46	SCHD	0.0115	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
53	VO	0.0115	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
74	QDF	0.0115	FlexShares Quality Dividend Index Fund	Dividend Equity	LONG
79	DIA	0.0115	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
12	PWV	0.0114	Invesco Large Cap Value ETF	Value Equity	LONG
13	RDVY	0.0114	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
58	SDOG	0.0114	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	LONG
60	DVY	0.0114	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
75	DGRW	0.0114	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
39	VLUE	0.0113	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
81	LGLV	0.0113	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	FLAT
1	XLI	0.0112	State Street Industrial Select Sector S...	Industrials	FLAT
6	FXR	0.0112	First Trust Industrials AlphaDEX	Industrials	LONG
17	REGL	0.0112	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
24	CFO	0.0112	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
40	VIS	0.0112	Vanguard Industrials ETF	Industrials	FLAT
42	FIDU	0.0112	Fidelity MSCI Industrials Index ETF	Industrials	FLAT

64	XMLV	0.0112	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
68	SMMV	0.0112	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
69	SPYD	0.0112	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	LONG
71	FTCS	0.0112	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
73	OUSA	0.0112	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
62	RPV	0.0111	Invesco S&P 500 Pure Value ETF	Value Equity	LONG
14	MOAT	0.0110	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
20	DHS	0.0110	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	LONG
88	OMFL	0.0110	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	LONG
11	FIW	0.0109	First Trust Water ETF	Utilities	LONG
80	USMV	0.0109	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
9	PHO	0.0108	Invesco Water Resources ETF	Utilities	FLAT
22	CDC	0.0108	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
61	RDIV	0.0108	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	LONG
84	PAVE	0.0108	Global X U.S. Infrastructure Developmen...	Industrials	FLAT
56	PEY	0.0107	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
63	SPHD	0.0107	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
15	FDL	0.0106	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
57	DIV	0.0106	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
3	ACWV	0.0105	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
65	XSLV	0.0105	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
29	LVHD	0.0104	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
59	SMDV	0.0104	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
10	CGW	0.0103	Invesco S&P Global Water Index ETF	Utilities	FLAT
43	HDV	0.0103	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
78	SPLV	0.0101	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	LONG
85	IYZ	0.0097	iShares U.S. Telecommunications ETF	Communication Services	FLAT
2	REM	0.0092	iShares Mortgage Real Estate ETF	Real Estate	FLAT

Cluster 3 – Developed ex-US Equity

Members: 61 **Var explained:** 0.898 **Signal:** LONG since 2026-04-10 (31 days, 3.5%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +47.7%)
- Latest sell_cross: 2026-03-05
- Current 252d return: +27.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-05)



Category breakdown

category	n_members	abs_weight
International Equity	19	0.3159
Developed ex-US Equity	9	0.1527
Global ex-US Equity	5	0.0849
Europe Equity	4	0.0674
International Dividend Equity	4	0.0660
Alternatives	4	0.0614
Eurozone Equity	3	0.0489
Commodity	1	0.0171
Global ex-US Small Cap Equity	1	0.0166
Germany Equity	1	0.0162
France Equity	1	0.0162
Italy Equity	1	0.0159
United Kingdom Equity	1	0.0158
Europe Financials	1	0.0158
Netherlands Equity	1	0.0157
Sweden Equity	1	0.0156
Spain Equity	1	0.0150
Austria Equity	1	0.0147
Belgium Equity	1	0.0146
Poland Equity	1	0.0135

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
VEA	0.971	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0172	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0172	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
41	ESGD	0.0172	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
51	IEFA	0.0172	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
52	SCHF	0.0172	Schwab International Equity ETF	Developed ex-US Equity	LONG
53	SPDW	0.0172	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
25	TLTD	0.0171	FlexShares Morningstar Developed Market...	International Equity	FLAT
40	GSIE	0.0171	Goldman Sachs ActiveBeta International ...	Commodity	FLAT
6	IEV	0.0170	iShares Europe ETF	Europe Equity	FLAT
16	DWM	0.0170	WisdomTree International Equity Fund	International Dividend E...	FLAT
19	PXF	0.0170	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
29	ACWX	0.0170	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
32	INTF	0.0170	iShares International Equity Factor ETF	International Equity	FLAT
46	IEUR	0.0170	iShares Core MSCI Europe ETF	Europe Equity	FLAT
47	IXUS	0.0170	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
48	VEU	0.0170	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
49	VGK	0.0170	Vanguard FTSEEurope ETF	Europe Equity	FLAT
50	VXUS	0.0170	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
30	CWI	0.0169	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
33	IQLT	0.0169	iShares MSCI Intl Quality Factor ETF	International Equity	FLAT
35	QEFA	0.0169	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
39	FNDF	0.0169	Schwab Fundamental International Equity...	International Equity	LONG
23	EFV	0.0168	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT
14	EZU	0.0167	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
18	IQDF	0.0167	FlexShares International Quality Divide...	International Equity	LONG
22	SCZ	0.0167	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG
24	FNDC	0.0167	Schwab Fundamental International Small ...	International Equity	LONG
26	JPIN	0.0167	JPMorgan Diversified Return Internation...	International Equity	FLAT
37	RODM	0.0167	Hartford Multifactor Developed Markets ...	International Equity	FLAT
38	VYMI	0.0167	Vanguard International High Dividend Yi...	International Dividend E...	LONG
44	SCHC	0.0167	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
56	HFXI	0.0167	NYLI FTSE International Equity Currency...	International Equity	LONG
20	EFG	0.0166	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
34	IVLU	0.0166	iShares MSCI Intl Value Factor ETF	International Equity	FLAT

36	FEZ	0.0166	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
45	VSS	0.0166	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	FLAT
4	FEP	0.0165	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
7	DLS	0.0165	WisdomTree International SmallCap Fund	International Equity	LONG
31	IMTM	0.0164	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
43	VIGI	0.0163	Vanguard International Dividend Appreci...	International Dividend E...	LONG
1	EWG	0.0162	iShares MSCI Germany Index Fund	Germany Equity	LONG
12	EWQ	0.0162	iShares MSCI France Index Fund	France Equity	FLAT
21	GWX	0.0162	State Street SPDR S&P International Sma...	International Equity	FLAT
5	VIDI	0.0160	Vident International Equity Strategy ETF	International Equity	LONG
59	IPKW	0.0160	Invesco International BuyBack Achievers...	International Equity	LONG
15	IDV	0.0159	iShares International Select Dividend E...	International Dividend E...	LONG
61	EWI	0.0159	iShares MSCI Italy ETF	Italy Equity	LONG
13	EWU	0.0158	iShares MSCI United Kingdom ETF	United Kingdom Equity	FLAT
17	EUFN	0.0158	iShares MSCI Europe Financials ETF	Europe Financials	FLAT
42	HDEF	0.0158	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
58	EWN	0.0157	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
10	HEZU	0.0156	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
27	DBEF	0.0156	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
28	HEFA	0.0156	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
55	EWD	0.0156	iShares MSCI Sweden ETF	Sweden Equity	LONG
8	HEDJ	0.0151	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
9	IHDG	0.0151	WisdomTree International Hedged Quality...	Alternatives	LONG
11	EWSP	0.0150	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EWAO	0.0147	iShares MSCI Austria ETF	Austria Equity	LONG
60	EWBK	0.0146	iShares MSCI Belgium ETF	Belgium Equity	FLAT
54	EPOL	0.0135	iShares MSCI Poland ETF	Poland Equity	FLAT

Cluster 4 – U.S. Small Cap Equity

Members: 52 **Var explained:** 0.901 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +41.4%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +29.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-15 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
U.S. Small Cap Equity	9	0.1792
U.S. Small Cap Value	6	0.1194
Consumer Discretionary	6	0.1050
Industrials	5	0.0871
U.S. Mid Cap Value	3	0.0597
U.S. Small Cap Growth	3	0.0595
Leveraged / Inverse	2	0.0386
U.S. Small Cap Fundamental Equity	1	0.0202
U.S. Small Cap ESG Equity	1	0.0200
U.S. Small/Mid Cap Fundamental Equity	1	0.0200
Mid Cap Value Equities	1	0.0200
U.S. Mid/Small Cap Equity	1	0.0199
U.S. Small Cap Multifactor Equity	1	0.0198
Small Cap Value Equities	1	0.0198
U.S. Mid Cap Equity	1	0.0198
Dividend Equity	1	0.0197
U.S. Small/Mid Cap Equity	1	0.0197
Value Equity	1	0.0195
U.S. Mid Cap Dividend Equity	1	0.0195
U.S. Small Cap Dividend Equity	1	0.0194
U.S. Equity	1	0.0193
U.S. Small Cap Industrials	1	0.0190

U.S. Small Cap Momentum Equity	1	0.0189
U.S. Micro Cap Equity	1	0.0186
U.S. Small Cap Consumer Discretionary	1	0.0184

Replication basket – IJR (cor=0.99) basket lead at lag 1: -0.015

ticker	rep_wt	name	category	etf_signal
IJR	0.978	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
31	FNDA	0.0202	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
13	VI00	0.0201	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
15	IJR	0.0201	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
16	SPSM	0.0201	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
40	VBR	0.0201	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
5	FYX	0.0200	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
7	NUSC	0.0200	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
8	PRFZ	0.0200	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
27	EZM	0.0200	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
4	FNX	0.0199	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
17	SCHA	0.0199	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
32	IWN	0.0199	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
33	IJJ	0.0199	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
34	IJS	0.0199	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
35	IVOV	0.0199	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
36	MDYV	0.0199	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
37	SLYV	0.0199	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
39	VTWV	0.0199	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
3	IWM	0.0198	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	FLAT
9	SMLF	0.0198	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	FLAT
10	IJT	0.0198	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
11	VIOG	0.0198	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
12	SLYG	0.0198	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
14	VTWO	0.0198	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	FLAT
6	EES	0.0198	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
38	VIOV	0.0198	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
41	RWJ	0.0198	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
45	RWK	0.0198	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
51	SDVY	0.0197	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	FLAT

52	ISMD	0.0197	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
28	PSC	0.0196	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
25	DON	0.0195	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
50	FYT	0.0195	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
24	DES	0.0194	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
44	RFV	0.0194	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
46	OMFS	0.0193	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
47	RZV	0.0192	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
42	PSCI	0.0190	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
18	FXD	0.0189	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
49	XSMO	0.0189	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
6	IWC	0.0186	iShares Microcap ETF	U.S. Micro Cap Equity	FLAT
23	RSPD	0.0185	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
48	PSCD	0.0184	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
20	FTXR	0.0183	First Trust Nasdaq Transportation ETF	Industrials	LONG
30	XTN	0.0180	State Street SPDR S&P Transportation ETF	Industrials	FLAT
1	XRT	0.0177	State Street SPDR S&P Retail ETF	Consumer Discretionary	FLAT
43	PKB	0.0176	Invesco Building & Construction ETF	Industrials	FLAT
22	IYT	0.0173	iShares U.S. Transportation ETF	Industrials	FLAT
19	PEJ	0.0172	Invesco Leisure and Entertainment ETF	Consumer Discretionary	LONG
29	XHB	0.0171	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
2	JETS	0.0159	U.S. Global Jets ETF	Industrials	FLAT
21	ITB	0.0157	iShares U.S. Home Construction ETF	Consumer Discretionary	FLAT

Cluster 5 – Aggregate Fixed Income

Members: 43 **Var explained:** 0.842 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-08 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +5.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-08)



Category breakdown

category	n_members	abs_weight
Fixed Income	13	0.3081
Aggregate Fixed Income	6	0.1478
Investment Grade Corporate Bond	6	0.1441
Mortgage-Backed Fixed Income	6	0.1326
Active Fixed Income	4	0.0943
Cash / Ultra-Short Fixed Income	3	0.0667
Target Maturity Bond	2	0.0422
International Fixed Income	2	0.0417
Agency Fixed Income	1	0.0227

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
BND	0.729	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

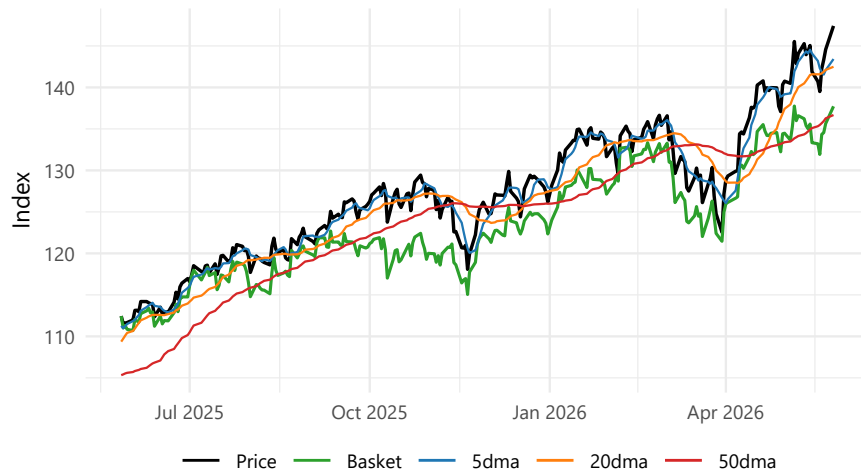
	ticker	PC_wt	name	category	etf_signal
13	BIV	0.0250	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
16	AGG	0.0249	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	FLAT
33	VGIT	0.0249	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
17	BND	0.0248	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
19	SCHZ	0.0248	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
29	IEI	0.0248	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
30	SPTI	0.0248	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
32	SCHR	0.0248	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
12	IUSB	0.0247	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	FLAT
20	SPAB	0.0247	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
1	IEF	0.0246	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
6	GVI	0.0246	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
11	ISTB	0.0245	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
9	SPIB	0.0243	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
10	IGIB	0.0243	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
14	BSV	0.0242	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
15	VCIT	0.0242	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
35	MBB	0.0242	iShares MBS ETF	Mortgage-Backed Fixed In...	FLAT
36	VMBS	0.0242	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	FLAT
5	FBND	0.0241	Fidelity Total Bond ETF	Active Fixed Income	FLAT
37	SPMB	0.0241	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	FLAT
24	IGSB	0.0240	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	LONG
7	AGGY	0.0239	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	FLAT
26	VCSH	0.0239	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
4	GTO	0.0238	Invesco Total Return Bond ETF	Active Fixed Income	FLAT
3	BOND	0.0237	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	FLAT
25	SLQD	0.0234	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
31	GOVT	0.0233	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
34	GNMA	0.0233	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
38	SHY	0.0228	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
28	AGZ	0.0227	iShares Agency Bond ETF	Agency Fixed Income	LONG
23	SPSB	0.0226	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
18	TOTL	0.0225	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
41	VGSH	0.0225	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
40	SCHO	0.0221	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
39	SPTS	0.0220	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
22	BSCQ	0.0214	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
42	BNDX	0.0210	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
8	IBDR	0.0208	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
43	IAGG	0.0207	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
27	CMBS	0.0186	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
2	LMBS	0.0181	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
21	LDUR	0.0171	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	FLAT

Cluster 6 – U.S. Mid Cap Equity

Members: 38 **Var explained:** 0.868 **Signal:** LONG since 2026-04-10 (31 days, 9.7%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +44.8%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +33.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-06)

**Category breakdown**

category	n_members	abs_weight
U.S. Mid Cap Growth	6	0.1645
U.S. Mid Cap Equity	5	0.1369
Industrials	4	0.0937
U.S. Small Cap Growth	3	0.0822
U.S. Momentum Equity	3	0.0762
Technology	3	0.0756
Growth Equity	2	0.0538
Momentum Equity	2	0.0521
Convertible Bond	2	0.0505
U.S. Extended Market Equity	1	0.0277
U.S. Small Cap Equity	1	0.0276

Unknown	1	0.0273
U.S. ESG Equity	1	0.0271
U.S. Mid Cap Momentum Equity	1	0.0265
U.S. Mid Cap Quality Equity	1	0.0265
U.S. Small Cap Momentum Equity	1	0.0260
IPO Equity	1	0.0259

Replication basket – IJH (cor=0.96) basket lead at lag 1: 0.003

ticker	rep_wt	name	category	etf_signal
IJH	0.989	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
19	IJK	0.0277	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
20	IVOG	0.0277	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
21	MDYG	0.0277	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
26	VXF	0.0277	Vanguard Extended Market ETF	U.S. Extended Market Equ...	FLAT
15	SCHM	0.0276	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
24	VBK	0.0276	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
29	VB	0.0276	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
13	IMCG	0.0274	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
14	JHMM	0.0274	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
25	ISCG	0.0274	iShares Morningstar Small-Cap Growth ETF	Growth Equity	FLAT
17	IWO	0.0273	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
18	MDY	0.0273	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
22	VTWG	0.0273	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
23	IVOO	0.0273	Vanguard S&P Mid-Cap 400 ETF		FLAT
27	IJH	0.0273	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
28	SPMD	0.0273	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
34	ETHO	0.0271	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
9	IWP	0.0270	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
12	VOT	0.0269	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
36	VFMO	0.0269	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
3	FTC	0.0267	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
4	PDP	0.0267	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
8	XMMO	0.0265	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
35	XMHQ	0.0265	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
7	RPG	0.0264	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
16	DWAS	0.0260	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT

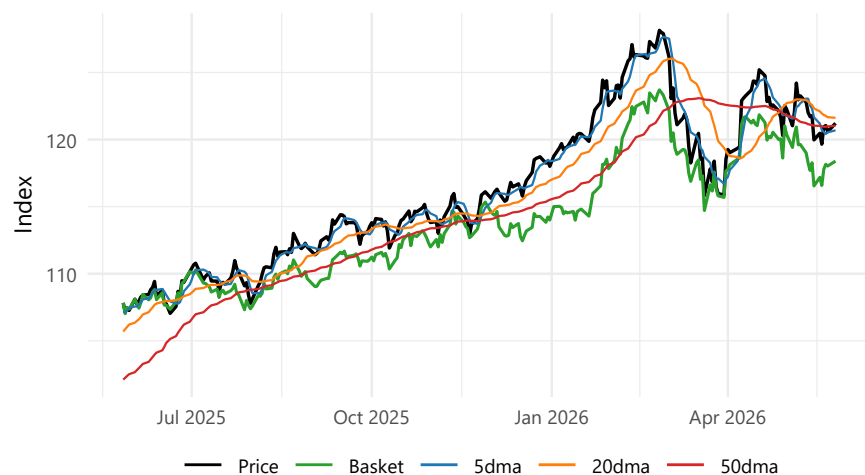
5	FPX	0.0259	First Trust US Equity Opportunities ETF	IPO Equity	LONG
6	CWB	0.0256	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
2	FV	0.0255	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
37	PRN	0.0254	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
10	ICVT	0.0249	iShares Convertible Bond ETF	Convertible Bond	LONG
1	ARKQ	0.0248	ARK Autonomous Technology & Robotics ETF	Technology	LONG
11	MTUM	0.0247	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
33	QMOM	0.0246	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	FLAT
38	PTF	0.0241	Invesco Dorsey Wright Technology Momen...	Technology	LONG
32	XAR	0.0236	State Street SPDR S&P Aerospace & Defen...	Industrials	LONG
30	PPA	0.0231	Invesco Aerospace & Defense ETF	Industrials	FLAT
31	ITA	0.0217	iShares U.S. Aerospace & Defense ETF	Industrials	LONG

Cluster 7 – Global Equity

Members: 30 **Var explained:** 0.602 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-05 – filter **PASS** (252d at date = +16.0%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +13.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-15 more recent than last effective buy 2026-05-05)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.0872
Asia-Pacific Equity	2	0.0805
International Equity	2	0.0787
International Dividend Equity	2	0.0787
Real Estate	2	0.0754
Global Equity	2	0.0753
Emerging Markets Fixed Income	2	0.0687
Currency	2	0.0657
Australia Equity	1	0.0399
Canada Equity	1	0.0379
Switzerland Equity	1	0.0355
South Africa Equity	1	0.0339
New Zealand Equity	1	0.0325
Latin America Equity	1	0.0322
Mexico Equity	1	0.0305
Malaysia Equity	1	0.0285
Brazil Equity	1	0.0272
Chile Equity	1	0.0270
Indonesia Equity	1	0.0248
Philippines Equity	1	0.0236
Vietnam Equity	1	0.0160

Replication basket – EWZ+IGF+INDA (cor=0.92) basket lead at lag 1: 0.009

ticker	rep_wt	name	category	etf_signal
EWZ	0.127	iShares MSCI Brazil ETF	Brazil Equity	FLAT
IGF	0.506	iShares Global Infrastructure ETF	Global Equity	LONG
INDA	0.245	iShares MSCI India ETF	India Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

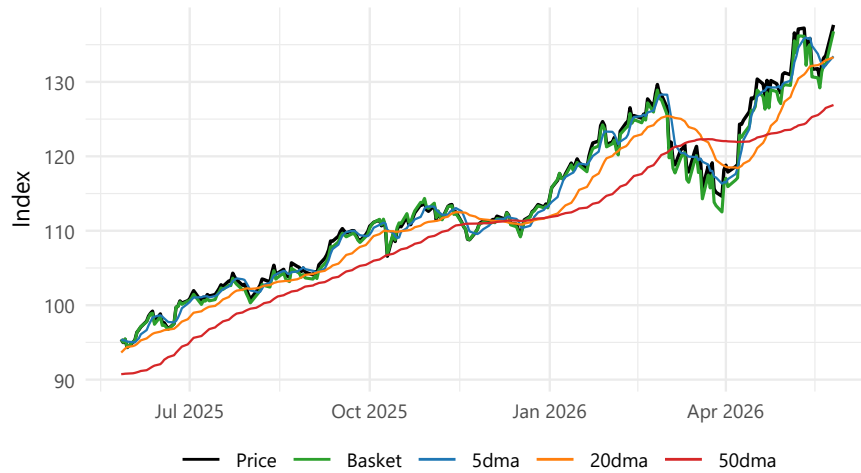
	ticker	PC_wt	name	category	etf_signal
8	EPP	0.0403	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
30	BBAX	0.0402	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
12	IDLV	0.0401	Invesco S&P International Developed Low...	International Equity	FLAT
5	EWA	0.0399	iShares MSCI Australia Index Fund	Australia Equity	FLAT
11	DWX	0.0395	State Street SPDR S&P International Div...	International Dividend E...	LONG
4	PID	0.0391	Invesco International Dividend Achiever...	International Dividend E...	LONG
13	EFAV	0.0385	iShares MSCI EAFE Min Vol Factor ETF	International Equity	LONG
9	NFRA	0.0383	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
14	VNQI	0.0382	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
7	EWC	0.0379	iShares MSCI Canada Index Fund	Canada Equity	LONG
3	RWX	0.0371	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
10	IGF	0.0370	iShares Global Infrastructure ETF	Global Equity	LONG
6	EWL	0.0355	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
22	FXA	0.0349	Invesco CurrencyShares Australian Dolla...	Currency	FLAT
17	EBND	0.0345	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	FLAT
18	EMLC	0.0342	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	FLAT
27	EZA	0.0339	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
26	ENZL	0.0325	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
21	ILF	0.0322	iShares Latin America 40 ETF	Latin America Equity	FLAT
28	FXC	0.0308	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
19	EWX	0.0305	iShares MSCI Mexico ETF	Mexico Equity	LONG
15	INDY	0.0295	iShares India 50 ETF	India Equity	FLAT
2	INDA	0.0291	iShares MSCI India ETF	India Equity	FLAT
16	EPI	0.0286	WisdomTree India Earnings Fund	India Equity	FLAT
25	EWM	0.0285	iShares MSCI Malaysia Index Fund	Malaysia Equity	FLAT
1	EWZ	0.0272	iShares MSCI Brazil ETF	Brazil Equity	FLAT
20	ECH	0.0270	iShares MSCI Chile ETF	Chile Equity	LONG
23	EIDO	0.0248	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
24	EPHE	0.0236	iShares MSCI Philippines ETF	Philippines Equity	FLAT
29	VNM	0.0160	VanEck Vietnam ETF	Vietnam Equity	FLAT

Cluster 8 – Emerging Markets Equity

Members: 25 **Var explained:** 0.874 **Signal:** LONG since 2026-04-10 (31 days, 10.3%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-03-05
- Current 252d return: +44.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-05)

**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7724
Asia Equity	2	0.0813
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0358
Global Dividend Equity	1	0.0346
South Korea Equity	1	0.0341

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.026

ticker	rep_wt	name	category	etf_signal
IEMG	0.929	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG
16	GEM	0.0423	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	FLAT
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	FLAT
18	XSOE	0.0419	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	LONG
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
15	EMGF	0.0418	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
11	EEMA	0.0413	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	FLAT
12	PXH	0.0410	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	FLAT
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	FLAT
19	EEMV	0.0404	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	FLAT
10	AIA	0.0401	iShares Asia 50 ETF	Asia Equity	LONG
7	DEM	0.0396	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0391	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	FLAT
4	FEM	0.0386	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0368	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0358	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
24	SDIV	0.0346	Global X SuperDividend ETF	Global Dividend Equity	FLAT
25	FEMS	0.0346	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	LONG

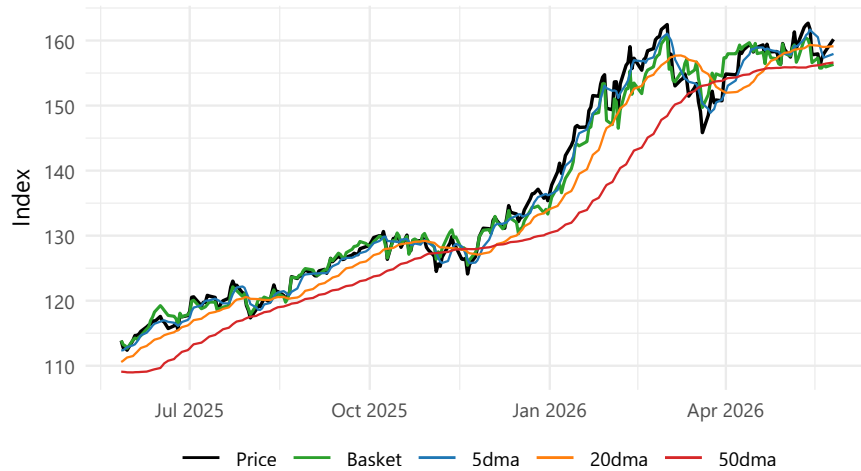
Cluster 9 – Materials

Members: 21 **Var explained:** 0.705 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +44.3%)

- Latest sell_cross: 2026-05-20
- Current 252d return: +44.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Materials	13	0.6531
MLP / Energy Infrastructure	4	0.1703
Energy	1	0.0501
Consumer Staples	1	0.0493
Norway Equity	1	0.0439
Industrial Metals	1	0.0333

Replication basket – AMLP+URA+GUNR (cor=0.96) basket lead at lag 1: 0.042

ticker	rep_wt	name	category	etf_signal
AMLP	0.114	Alerian MLP ETF	MLP / Energy Infrastr...	LONG
URA	0.081	Global X Uranium ETF	Materials	FLAT
GUNR	0.773	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

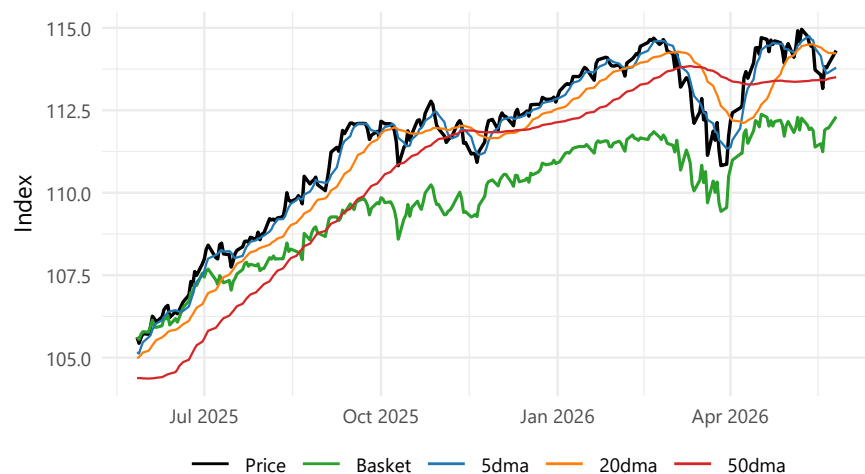
	ticker	PC_wt	name	category	etf_signal
15	GNR	0.0549	State Street SPDR S&P Global Natural Re...	Materials	FLAT
14	GUNR	0.0542	FlexShares Global Upstream Natural Reso...	Materials	FLAT
10	MXI	0.0528	iShares Global Materials ETF	Materials	FLAT
11	IYM	0.0528	iShares U.S. Basic Materials ETF	Materials	LONG
21	NANR	0.0523	State Street SPDR S&P North American Na...	Materials	FLAT
13	VAW	0.0513	Vanguard Materials ETF	Materials	FLAT
16	PICK	0.0508	iShares MSCI Global Select Metals & Min...	Materials	FLAT
12	RSPM	0.0506	Invesco S&P 500 Equal Weight Materials ...	Materials	FLAT
1	XLB	0.0505	State Street Materials Select Sector SP...	Materials	FLAT
20	IGE	0.0501	iShares North American Natural Resource...	Energy	LONG
4	XME	0.0498	State Street SPDR S&P Metals & Mining E...	Materials	LONG
17	SLX	0.0496	VanEck Steel ETF	Materials	LONG
6	MOO	0.0493	VanEck Agribusiness ETF	Consumer Staples	FLAT
3	COPX	0.0471	Global X Copper Miners ETF	Materials	LONG
5	EMLP	0.0441	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
9	MLPX	0.0439	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
18	NORW	0.0439	Global X MSCI Norway ETF	Norway Equity	LONG
7	AMLP	0.0417	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
8	MLPA	0.0406	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
2	URA	0.0365	Global X Uranium ETF	Materials	FLAT
19	DBB	0.0333	Invesco DB Base Metals Fund	Industrial Metals	LONG

Cluster 10 – High Yield Bond

Members: 21 **Var explained:** 0.766 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +10.3%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +8.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-18 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5154
Preferred Stock	7	0.3063
Emerging Markets Fixed Income	4	0.1783

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.049

ticker	rep_wt	name	category	etf_signal
HYG	0.916	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	JNK	0.0526	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
1	HYG	0.0525	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	FLAT
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	FLAT
18	BBHY	0.0522	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	FLAT

14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
17	FALN	0.0516	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
15	ANGL	0.0515	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0480	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0473	First Trust Tactical High Yield ETF	High Yield Bond	FLAT
2	EMB	0.0467	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWO	0.0466	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	LONG
11	PFF	0.0463	VanEck Preferred Securities ex Financia...	Preferred Stock	FLAT
20	PCY	0.0457	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0433	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
8	VRP	0.0369	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 11 – Healthcare

Members: 17 **Var explained:** 0.762 **Signal:** LONG since 2026-05-06 (13 days, 0.3%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.9%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +18.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.004

ticker	rep_wt	name	category	etf_signal
VHT	1.102	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0653	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0652	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0645	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0639	Invesco S&P 500 Equal Weight Health Car...	Healthcare	LONG
13	IYH	0.0637	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0623	iShares Global Healthcare ETF	Healthcare	LONG

1	XLV	0.0620	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0604	iShares Biotechnology ETF	Healthcare	FLAT
6	FBT	0.0596	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	LONG
12	IHI	0.0572	iShares U.S. Medical Devices ETF	Healthcare	FLAT
5	PTH	0.0544	Invesco Dorsey Wright Healthcare Moment...	Healthcare	FLAT
17	PPH	0.0543	VanEck Pharmaceutical ETF	Healthcare	LONG
4	XBI	0.0535	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0510	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0470	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 12 – China Equity

Members: 15 **Var explained:** 0.803 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +13.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)



Category breakdown

category	n_members	abs_weight
China Equity	13	0.8767
Emerging Markets Equity	1	0.0642
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.010

ticker	rep_wt	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

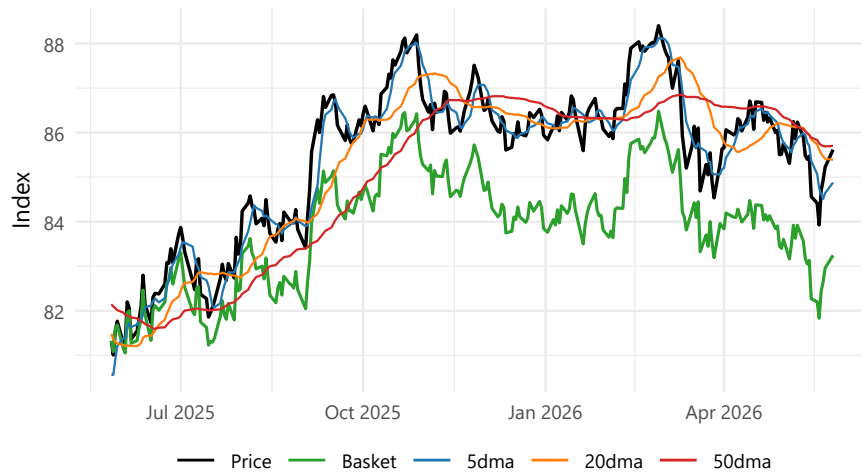
	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0736	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0735	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0729	iShares MSCI China ETF	China Equity	FLAT
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	LONG
7	CHIQ	0.0706	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0679	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	FLAT
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	FLAT
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0642	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0595	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
14	CNXT	0.0595	VanEck ChiNext Innovators ETF	China Equity	LONG
12	EWK	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
15	KURE	0.0587	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 13 – Fixed Income

Members: 15 **Var explained:** 0.928 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-02-10 – filter **PASS** (252d at date = +5.7%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +6.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-02-10)



Category breakdown

category	n_members	abs_weight
Investment Grade Corporate Bond	7	0.4706
Fixed Income	6	0.4009
Active Fixed Income	1	0.0665
Taxable Municipal Bond	1	0.0619

Replication basket – TLT (cor=0.97) basket lead at lag 1: -0.041

ticker	rep_wt	name	category	etf_signal
TLT	0.694	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
9	BLV	0.0687	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
7	SPLB	0.0678	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	FLAT
8	IGLB	0.0678	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	FLAT
10	VCLT	0.0676	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
2	LQD	0.0675	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	FLAT
11	USIG	0.0672	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	FLAT
6	QLTA	0.0671	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	FLAT
12	TLH	0.0667	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
14	SPTL	0.0667	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
15	VGLT	0.0666	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
3	FLCO	0.0665	Franklin Investment Grade Corporate ETF	Active Fixed Income	FLAT
1	TLT	0.0664	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
5	CORP	0.0657	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
13	EDV	0.0657	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
4	BAB	0.0619	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT

Cluster 14 – Financials

Members: 14 **Var explained:** 0.878 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +12.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.938
Global Equity	1	0.062

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.014

ticker	rep_wt	name	category	etf_signal
XLF	1.097	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0754	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0754	Fidelity MSCI Financials Index ETF	Financials	LONG
5	FXO	0.0751	First Trust Financials AlphaDEX	Financials	FLAT
7	IYF	0.0751	iShares U.S. Financial ETF	Financials	FLAT
8	IYG	0.0747	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0740	State Street Financial Select Sector SP...	Financials	LONG

9	KBWB	0.0732	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0711	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0708	State Street SPDR S&P Capital Markets E...	Financials	LONG
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	LONG
13	IAI	0.0700	iShares U.S. Broker-Dealers & Securitie...	Financials	LONG
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0640	State Street SPDR S&P Insurance ETF	Financials	LONG
12	PSP	0.0620	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.962 **Signal:** LONG since 2026-04-10 (31 days, 5.0%)

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-03-10
- Current 252d return: +16.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-10 > last sell at 2026-03-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: 0.001

ticker	rep_wt	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

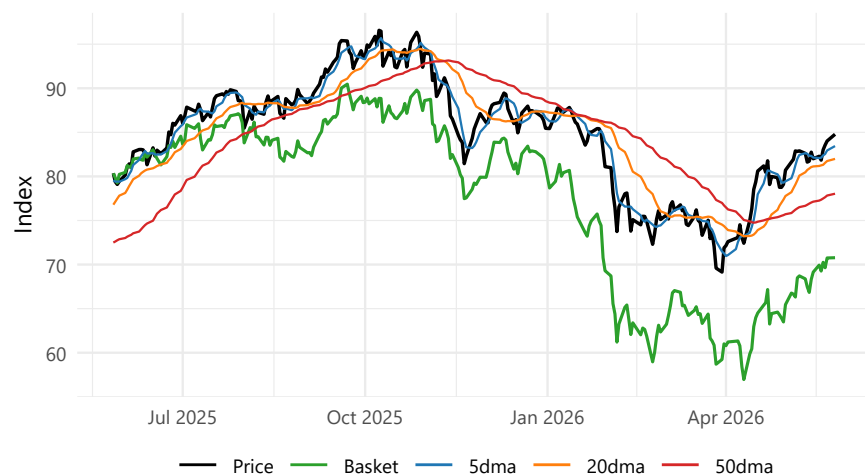
	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	FLAT
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	LONG
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	LONG
10	USRT	0.0723	iShares Core U.S. REIT ETF	Real Estate	LONG
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	LONG
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	FLAT
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	LONG
7	REET	0.0715	iShares Global REIT ETF	Real Estate	LONG
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	FLAT
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	LONG
14	NURE	0.0689	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	FLAT

Cluster 16 – Technology

Members: 14 **Var explained:** 0.820 **Signal:** LONG since 2026-04-15 (28 days, 8.0%)

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +17.1%)
- Latest sell_cross: 2026-03-24
- Current 252d return: +6.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-15 > last sell at 2026-03-24)



Category breakdown

category	n_members	abs_weight
Technology	10	0.7086
Financials	2	0.1451
IPO Equity	1	0.0739
Consumer Discretionary	1	0.0724

Replication basket – IGV (cor=0.91) basket lead at lag 1: 0.048

ticker	rep_wt	name	category	etf_signal
IGV	0.971	iShares Expanded Tech-Software Sector...	Technology	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	FDN	0.0752	First Trust DJ Internet Index Fund	Technology	LONG
2	SKYY	0.0751	First Trust Cloud Computing ETF	Technology	LONG
6	ARKW	0.0749	ARK Next Generation Internet ETF	Technology	LONG
13	FINX	0.0748	Global X FinTech ETF	Financials	FLAT

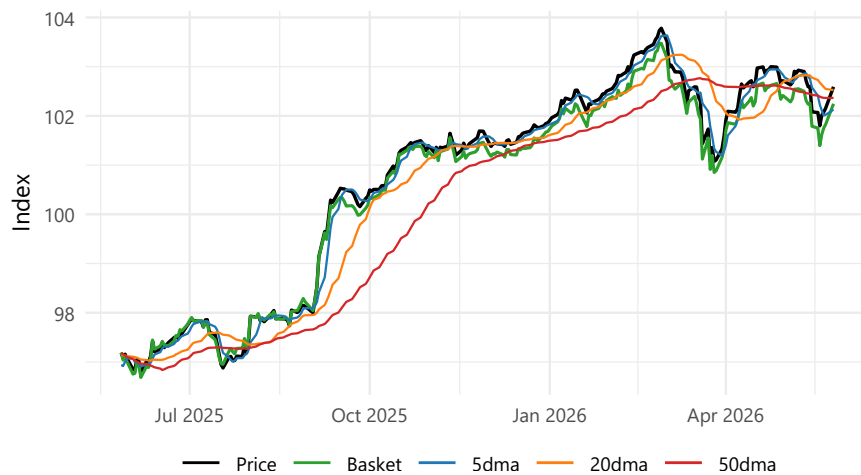
10	PNQI	0.0745	Invesco Nasdaq Internet ETF	Technology	FLAT
9	IPO	0.0739	Renaissance IPO ETF	IPO Equity	LONG
5	ARKK	0.0733	ARK Innovation ETF	Technology	FLAT
1	IGV	0.0729	iShares Expanded Tech-Software Sector E...	Technology	LONG
7	IBUY	0.0724	Amplify Online Retail ETF	Consumer Discretionary	FLAT
8	CIBR	0.0704	First Trust NASDAQ Cybersecurity ETF	Technology	LONG
12	IPAY	0.0702	Amplify Digital Payments ETF	Financials	FLAT
4	HACK	0.0698	Amplify Cybersecurity ETF	Technology	LONG
3	BLOK	0.0655	Amplify Blockchain Technology ETF	Technology	LONG
14	BLCN	0.0571	Siren NexGen Economy ETF	Technology	LONG

Cluster 17 – Municipal Bond

Members: 13 **Var explained:** 0.819 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-14 – filter **PASS** (252d at date = +6.9%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +6.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-15 more recent than last effective buy 2026-04-14)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.180

ticker	rep_wt	name	category	etf_signal
MUB	0.905	iShares National Muni Bond ETF	Municipal Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

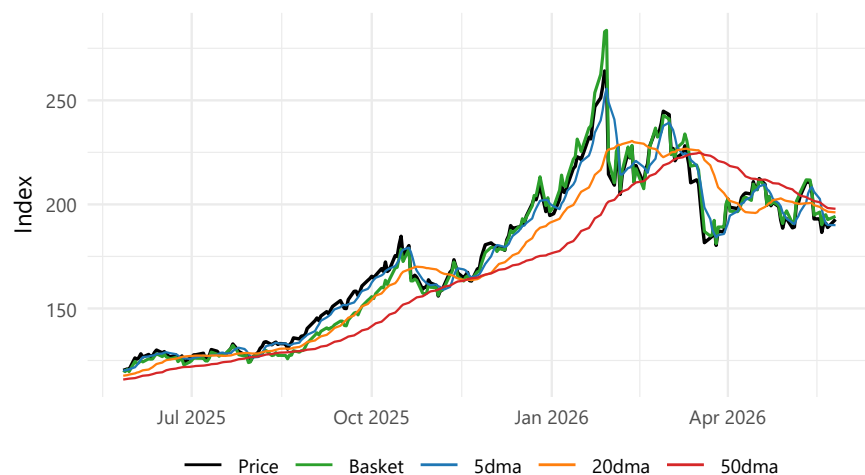
	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	FLAT
6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	FLAT
1	FMB	0.0795	First Trust Managed Municipal ETF	Municipal Bond	FLAT
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	FLAT
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	FLAT
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0747	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0695	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 18 – Gold

Members: 12 **Var explained:** 0.858 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +61.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



Category breakdown

category	n_members	abs_weight
Materials	5	0.4157
Gold	3	0.2501
Precious Metals	2	0.1725
Silver	2	0.1616

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.012

ticker	rep_wt	name	category	etf_signal
GLD	0.046	SPDR Gold Shares	Gold	FLAT
IAU	0.723	iShares Gold Trust Shares	Gold	FLAT
SLV	0.411	iShares Silver Trust	Silver	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0866	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0859	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT

1	GDX	0.0840	VanEck Gold Miners ETF	Materials	FLAT
2	GDXJ	0.0840	VanEck Junior Gold Miners ETF	Materials	FLAT
3	GLD	0.0834	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0828	Global X Silver Miners ETF	Materials	FLAT
5	SILJ	0.0817	Amplify Junior Silver Miners ETF	Materials	FLAT
4	SLV	0.0809	iShares Silver Trust	Silver	FLAT
8	SIVR	0.0807	abrdn Physical Silver Shares ETF	Silver	FLAT

Cluster 19 – Consumer Staples

Members: 11 **Var explained:** 0.746 **Signal:** LONG since 2026-05-22 (1 days, -0.8%)

Signal logic:

- Latest buy_cross: 2026-05-22 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +7.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-22 > last sell at 2026-05-13)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5729
Utilities	4	0.3525
U.S. Small Cap Consumer Staples	1	0.0746

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
XLU	0.365	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.587	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0974	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0948	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
3	KXI	0.0945	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0891	iShares U.S. Utilities ETF	Utilities	FLAT
9	FUTY	0.0880	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
8	VPU	0.0879	Vanguard Utilities ETF	Utilities	FLAT
2	XLU	0.0875	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0746	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	FLAT

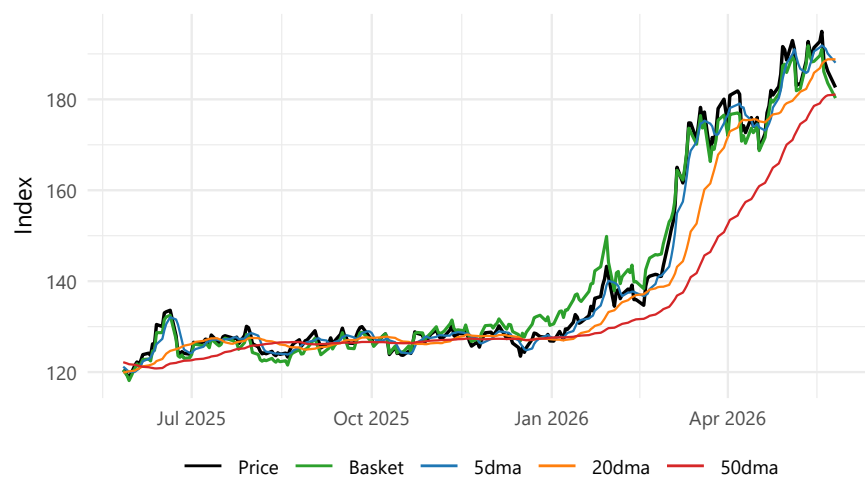
Cluster 20 – Broad Commodities

Members: 10 **Var explained:** 0.841 **Signal:** LONG since 2026-01-08 (94 days, 41.0%)

Signal logic:

- Latest buy_cross: 2026-01-08 – filter **PASS** (252d at date = +2.8%)
- Latest sell_cross: 2025-12-15
- Current 252d return: +51.4% – filter **ALLOWS** new entries today

- Decision: **LONG** (effective buy at 2026-01-08 > last sell at 2025-12-15)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6371
Oil	2	0.2056
Energy Commodities	1	0.1053
Agricultural Commodities	1	0.0520

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.074

ticker	rep_wt	name	category	etf_signal
PDBC	1.128	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	DBC	0.1091	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
6	GSG	0.1091	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT

8	PDBC	0.1090	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
9	COMT	0.1089	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
10	DBE	0.1053	Invesco DB Energy Fund	Energy Commodities	FLAT
1	USO	0.1034	United States Oil Fund	Oil	LONG
5	DBO	0.1022	Invesco DB Oil Fund	Oil	FLAT
4	FTGC	0.1005	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT
7	DJP	0.1005	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
3	DBA	0.0520	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 Var explained: 0.908 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +4.1%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +4.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-18 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
VTIP	0.276	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.755	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1033	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	FLAT
6	TIPX	0.1032	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1028	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
9	STIP	0.0999	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0998	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0997	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0995	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0886	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.887 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +42.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-04-29 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7799
Asia-Pacific Equity	2	0.2201

Replication basket – EWJ (cor=0.96) basket lead at lag 1: -0.036

ticker	rep_wt	name	category	etf_signal
EWJ	0.908	iShares MSCI Japan Index Fund	Japan Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1141	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1141	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
9	HEWJ	0.1116	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
6	DBJP	0.1115	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1107	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT

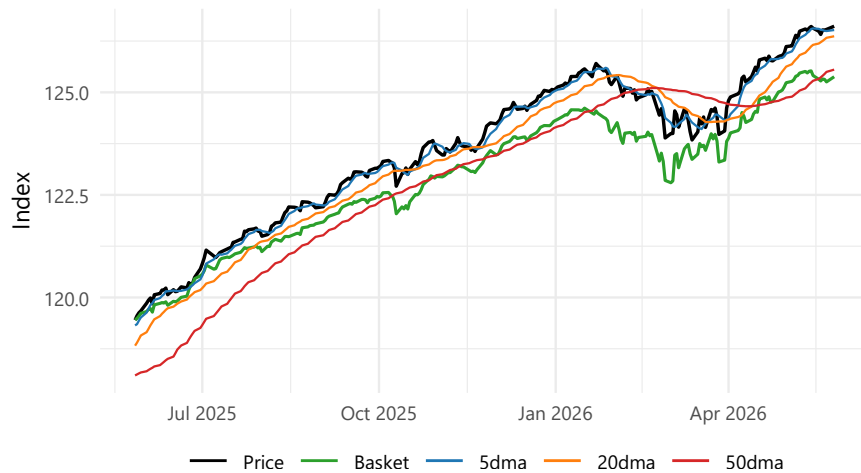
2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1088	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1082	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.559 **Signal:** LONG since 2026-04-07 (34 days, 1.3%)

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-02-03
- Current 252d return: +6.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-07 > last sell at 2026-02-03)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3630
Interest Rate Hedged Fixed Income	3	0.3245

Floating Rate Fixed Income 3 0.3125

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.048

ticker	rep_wt	name	category	etf_signal
FLOT	0.293	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.209	Invesco Senior Loan ETF	Bank Loan / Floating ...	FLAT
SRLN	0.311	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.127	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

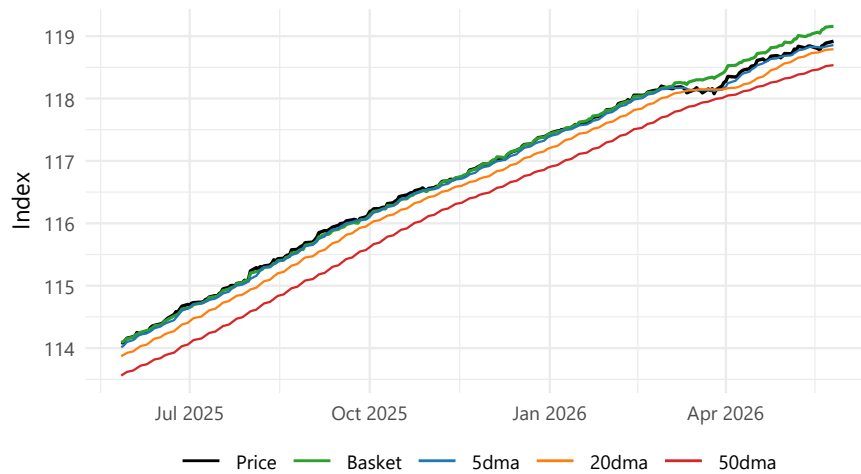
	ticker	PC_wt	name	category	etf_signal
2	SRLN	0.1249	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG
3	BKLN	0.1213	Invesco Senior Loan ETF	Bank Loan / Floating Rate	FLAT
1	FTSL	0.1168	First Trust Senior Loan Fund	Bank Loan / Floating Rate	FLAT
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	LONG
7	FLOT	0.1082	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1058	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	LONG
4	IGHG	0.1042	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	LONG
8	FLRN	0.1036	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1007	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 **Var explained:** 0.453 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2022-10-31 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-03-24
- Current 252d return: +4.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-03-24 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.106

ticker	rep_wt	name	category	etf_signal
SHV	0.347	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.175	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	LONG
GBIL	0.118	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	LONG
ICSH	0.418	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	GSY	0.1337	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	FLAT
3	FTSM	0.1288	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
7	SHV	0.1265	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1245	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT

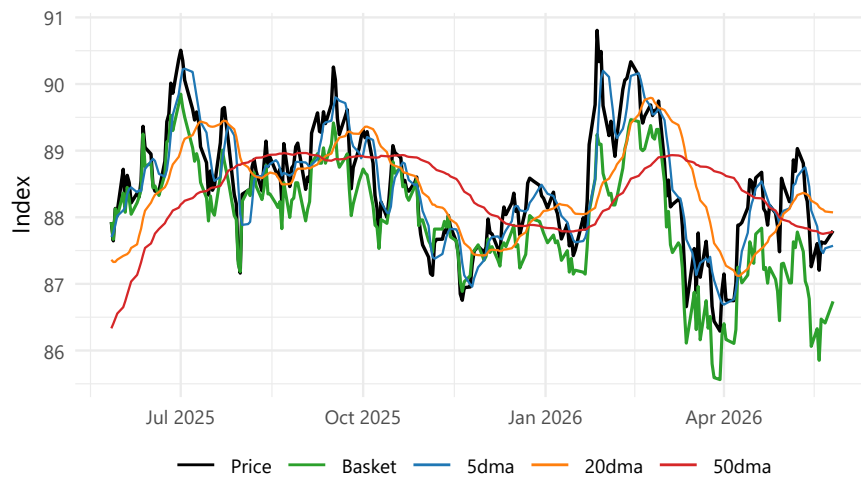
4	NEAR	0.1091	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1065	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	LONG
1	VNLA	0.1035	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.0883	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
9	GBIL	0.0790	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	LONG

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-30 – filter **PASS** (252d at date = +0.3%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +0.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-18 more recent than last effective buy 2026-04-30)



Category breakdown

category	n_members	abs_weight
Currency	4	0.641

International Fixed Income	2	0.359
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Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.027

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1789	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1701	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1648	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1610	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1451	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Currency

Members: 5 **Var explained:** 0.290 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-20 – filter **BLOCKED** (252d at date = -13.9%; needs > 0)
- Latest effective buy: 2025-04-04 – delayed re-arm from blocked cross (252d at fire = +0.9%)
- Latest sell_cross: 2026-04-09
- Current 252d return: -14.8% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-04-09 more recent than last effective buy 2025-04-04)



Category breakdown

category	n_members	abs_weight
Leveraged / Inverse	1	0.2956
Currency	1	0.2609
Turkey Equity	1	0.2410
Platinum	1	0.1402
Natural Gas	1	0.0624

Replication basket – SQQQ+UUP (cor=0.83) basket lead at lag 1: 0.009

ticker	rep_wt	name	category	etf_signal
SQQQ	0.339	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.517	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	SQQQ	0.2956	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
5	UUP	0.2609	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

4	TUR	0.2410	iShares MSCI Turkey ETF	Turkey Equity	FLAT
1	PPLT	0.1402	abrdn Physical Platinum Shares ETF	Platinum	FLAT
3	UNG	0.0624	United States Natural Gas Fund LP	Natural Gas	FLAT

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
UUP	cluster_26	Currency	2026-05-18	5	0.2%	0.00	1.00
GBIL	cluster_24	Cash / Ultra-Short Fixe...	2022-08-18	944	17.6%	0.00	1.00
MINT	cluster_24	Cash / Ultra-Short Fixe...	2023-01-30	832	18.9%	0.00	1.00
BIL	cluster_24	Cash / Ultra-Short Fixe...	2022-05-19	1006	18.2%	0.00	1.00
VNLA	cluster_24	Cash / Ultra-Short Fixe...	2026-04-02	36	0.7%	0.00	1.00
EWJV	cluster_22	Japan Equity	2026-05-06	13	2.5%	0.00	1.00
DXJ	cluster_22	Japan Equity	2026-05-01	16	6.1%	0.01	0.99
HEWJ	cluster_22	Japan Equity	2026-04-08	33	4.9%	0.01	0.99
DBJP	cluster_22	Japan Equity	2026-04-08	33	4.6%	0.01	0.99
EWJ	cluster_22	Japan Equity	2026-04-13	30	5.1%	0.12	0.88
BBJP	cluster_22	Japan Equity	2026-04-30	17	4.1%	0.12	0.88
VPL	cluster_22	Japan Equity	2026-04-10	31	11.1%	0.16	0.84
QQEW	cluster_1	U.S. Large Cap Equity	2026-04-16	27	12.9%	0.00	1.00
QUS	cluster_1	U.S. Large Cap Equity	2026-04-14	29	4.8%	0.00	1.00
QTEC	cluster_1	U.S. Large Cap Equity	2026-04-10	31	38.8%	0.00	1.00
SPHQ	cluster_1	U.S. Large Cap Equity	2026-04-10	31	8.7%	0.00	1.00
EWS	cluster_1	U.S. Large Cap Equity	2026-04-07	34	3.6%	0.00	1.00
NXTG	cluster_1	U.S. Large Cap Equity	2026-04-08	33	29.9%	0.00	1.00
XSD	cluster_1	U.S. Large Cap Equity	2026-04-09	32	73.6%	0.00	1.00
FTEC	cluster_1	U.S. Large Cap Equity	2026-04-10	31	27.2%	0.00	1.00
VGT	cluster_1	U.S. Large Cap Equity	2026-04-10	31	27.1%	0.00	1.00
XLK	cluster_1	U.S. Large Cap Equity	2026-04-10	31	29.8%	0.00	1.00
PTNQ	cluster_1	U.S. Large Cap Equity	2026-04-17	26	12.7%	0.00	1.00
XNTK	cluster_1	U.S. Large Cap Equity	2026-04-10	31	30.4%	0.00	1.00
PRNT	cluster_1	U.S. Large Cap Equity	2026-04-16	27	15.4%	0.00	1.00
QTUM	cluster_1	U.S. Large Cap Equity	2026-04-10	31	35.9%	0.00	1.00
XT	cluster_1	U.S. Large Cap Equity	2026-04-13	30	15.2%	0.01	0.99
IOO	cluster_1	U.S. Large Cap Equity	2026-04-13	30	10.7%	0.01	0.99
AIQ	cluster_1	U.S. Large Cap Equity	2026-04-13	30	28.9%	0.01	0.99
IXN	cluster_1	U.S. Large Cap Equity	2026-04-10	31	29.5%	0.01	0.99
PSCT	cluster_1	U.S. Large Cap Equity	2026-04-07	34	39.9%	0.01	0.99
PTLC	cluster_1	U.S. Large Cap Equity	2026-04-20	25	5.8%	0.01	0.99
QQQ	cluster_1	U.S. Large Cap Equity	2026-04-10	31	19.5%	0.01	0.99
IGM	cluster_1	U.S. Large Cap Equity	2026-04-09	32	26.6%	0.01	0.99
IYW	cluster_1	U.S. Large Cap Equity	2026-04-10	31	27.4%	0.01	0.99
QUAL	cluster_1	U.S. Large Cap Equity	2026-04-14	29	6.2%	0.02	0.98
LRGF	cluster_1	U.S. Large Cap Equity	2026-04-13	30	8.9%	0.02	0.98

ESGU	cluster_1	U.S. Large Cap Equity	2026-04-13	30	10.0%	0.02	0.98
MGC	cluster_1	U.S. Large Cap Equity	2026-04-13	30	10.4%	0.03	0.97
OEF	cluster_1	U.S. Large Cap Equity	2026-04-13	30	11.1%	0.03	0.97
SUSA	cluster_1	U.S. Large Cap Equity	2026-04-13	30	10.0%	0.03	0.97
EPS	cluster_1	U.S. Large Cap Equity	2026-04-10	31	10.0%	0.03	0.97
IWL	cluster_1	U.S. Large Cap Equity	2026-04-13	30	10.0%	0.03	0.97
SOXX	cluster_1	U.S. Large Cap Equity	2026-04-08	33	53.9%	0.03	0.97
NULG	cluster_1	U.S. Large Cap Equity	2026-04-10	31	16.7%	0.04	0.96
VV	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.5%	0.04	0.96
VOO	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.4%	0.05	0.95
IVV	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.4%	0.05	0.95
SMH	cluster_1	U.S. Large Cap Equity	2026-04-08	33	42.4%	0.05	0.95
SPY	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.4%	0.05	0.95
XLG	cluster_1	U.S. Large Cap Equity	2026-04-13	30	10.7%	0.05	0.95
FTXL	cluster_1	U.S. Large Cap Equity	2026-04-07	34	68.1%	0.05	0.95
TDIV	cluster_1	U.S. Large Cap Equity	2026-04-13	30	18.8%	0.05	0.95
ILCB	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.5%	0.05	0.95
URTH	cluster_1	U.S. Large Cap Equity	2026-04-13	30	7.4%	0.06	0.94
IWY	cluster_1	U.S. Large Cap Equity	2026-04-14	29	9.6%	0.06	0.94
SCHX	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.2%	0.06	0.94
VONE	cluster_1	U.S. Large Cap Equity	2026-04-13	30	8.9%	0.06	0.94
FXL	cluster_1	U.S. Large Cap Equity	2026-04-09	32	23.7%	0.07	0.93
SPYX	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.5%	0.07	0.93
IYY	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.1%	0.07	0.93
MGK	cluster_1	U.S. Large Cap Equity	2026-04-13	30	14.0%	0.07	0.93
IWB	cluster_1	U.S. Large Cap Equity	2026-04-13	30	8.9%	0.07	0.93
SPTM	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.0%	0.08	0.92
GSLC	cluster_1	U.S. Large Cap Equity	2026-04-13	30	8.1%	0.08	0.92
SCHB	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.1%	0.09	0.91
VTI	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.1%	0.09	0.91
ITOT	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.2%	0.09	0.91
IWV	cluster_1	U.S. Large Cap Equity	2026-04-13	30	8.9%	0.09	0.91
IWF	cluster_1	U.S. Large Cap Equity	2026-04-14	29	9.1%	0.09	0.91
VONG	cluster_1	U.S. Large Cap Equity	2026-04-14	29	9.1%	0.09	0.91
PSI	cluster_1	U.S. Large Cap Equity	2026-04-07	34	62.4%	0.09	0.91
VTHR	cluster_1	U.S. Large Cap Equity	2026-04-13	30	9.0%	0.10	0.90
VUG	cluster_1	U.S. Large Cap Equity	2026-04-13	30	13.2%	0.12	0.88
FPXI	cluster_1	U.S. Large Cap Equity	2026-04-10	31	13.2%	0.12	0.88
ONEQ	cluster_1	U.S. Large Cap Equity	2026-04-13	30	15.0%	0.12	0.88
DSI	cluster_1	U.S. Large Cap Equity	2026-04-14	29	9.7%	0.13	0.87
JHML	cluster_1	U.S. Large Cap Equity	2026-04-13	30	7.5%	0.14	0.86
ACWI	cluster_1	U.S. Large Cap Equity	2026-04-13	30	7.8%	0.15	0.85
CRBN	cluster_1	U.S. Large Cap Equity	2026-04-13	30	7.8%	0.16	0.84
SPYG	cluster_1	U.S. Large Cap Equity	2026-04-13	30	13.2%	0.17	0.83
IVW	cluster_1	U.S. Large Cap Equity	2026-04-13	30	13.2%	0.17	0.83

VOOG	cluster_1	U.S. Large Cap Equity	2026-04-13	30	13.2%	0.17	0.83
TILT	cluster_1	U.S. Large Cap Equity	2026-04-13	30	7.7%	0.18	0.82
CATH	cluster_1	U.S. Large Cap Equity	2026-04-13	30	8.6%	0.18	0.82
VT	cluster_1	U.S. Large Cap Equity	2026-04-13	30	7.5%	0.19	0.81
SCHG	cluster_1	U.S. Large Cap Equity	2026-04-13	30	11.7%	0.19	0.81
SPGM	cluster_1	U.S. Large Cap Equity	2026-04-10	31	8.5%	0.19	0.81
IUSG	cluster_1	U.S. Large Cap Equity	2026-04-13	30	12.9%	0.20	0.80
ILCG	cluster_1	U.S. Large Cap Equity	2026-04-13	30	13.4%	0.20	0.80
PWB	cluster_1	U.S. Large Cap Equity	2026-04-10	31	16.4%	0.23	0.77
AOA	cluster_1	U.S. Large Cap Equity	2026-04-13	30	5.9%	0.24	0.76
ROBO	cluster_1	U.S. Large Cap Equity	2026-04-14	29	18.1%	0.26	0.74
AOR	cluster_1	U.S. Large Cap Equity	2026-04-13	30	4.4%	0.34	0.66
QAI	cluster_1	U.S. Large Cap Equity	2026-04-08	33	4.4%	0.45	0.55
SPHB	cluster_1	U.S. Large Cap Equity	2026-04-10	31	19.4%	0.50	0.50
AOM	cluster_1	U.S. Large Cap Equity	2026-04-13	30	2.9%	0.51	0.49
AOK	cluster_1	U.S. Large Cap Equity	2026-04-13	30	2.1%	0.64	0.36
BOTZ	cluster_1	U.S. Large Cap Equity	2026-04-16	27	11.4%	0.70	0.30
IYC	cluster_1	U.S. Large Cap Equity	2026-04-14	29	-0.2%	0.89	0.03
FCOM	cluster_1	U.S. Large Cap Equity	2026-04-13	30	3.3%	0.99	0.01
VOX	cluster_1	U.S. Large Cap Equity	2026-04-13	30	3.5%	0.99	0.01
XLV	cluster_11	Healthcare	2026-05-21	2	0.2%	0.00	1.00
IYH	cluster_11	Healthcare	2026-05-20	3	1.0%	0.00	1.00
FHLC	cluster_11	Healthcare	2026-05-19	4	1.2%	0.00	1.00
VHT	cluster_11	Healthcare	2026-05-19	4	1.2%	0.00	1.00
PPH	cluster_11	Healthcare	2026-05-21	2	0.3%	0.00	1.00
IXJ	cluster_11	Healthcare	2026-05-22	1	-0.9%	0.00	1.00
RSPH	cluster_11	Healthcare	2026-05-21	2	-0.1%	0.04	0.95
FXH	cluster_11	Healthcare	2026-04-10	31	2.1%	0.09	0.91
IHF	cluster_11	Healthcare	2026-05-08	11	-0.4%	0.28	0.72
XHE	cluster_11	Healthcare	2026-05-22	1	0.1%	0.21	0.56
PSCH	cluster_11	Healthcare	2026-04-13	30	4.6%	0.98	0.02
VIGI	cluster_3	Developed ex-US Equity	2026-04-14	29	1.9%	0.00	1.00
HDEF	cluster_3	Developed ex-US Equity	2026-04-07	34	1.6%	0.00	1.00
IDV	cluster_3	Developed ex-US Equity	2026-04-06	35	4.9%	0.01	0.99
VIDI	cluster_3	Developed ex-US Equity	2026-04-09	32	9.2%	0.01	0.99
FNDF	cluster_3	Developed ex-US Equity	2026-04-08	33	5.8%	0.02	0.98
PXF	cluster_3	Developed ex-US Equity	2026-04-08	33	5.9%	0.02	0.98
EWN	cluster_3	Developed ex-US Equity	2026-04-10	31	7.8%	0.02	0.98
HFXI	cluster_3	Developed ex-US Equity	2026-04-09	32	6.9%	0.03	0.97
IHDG	cluster_3	Developed ex-US Equity	2026-05-06	13	0.7%	0.04	0.96
VYMI	cluster_3	Developed ex-US Equity	2026-04-09	32	2.5%	0.08	0.92
SCHF	cluster_3	Developed ex-US Equity	2026-04-10	31	6.2%	0.08	0.92
EWG	cluster_3	Developed ex-US Equity	2026-04-14	29	3.7%	0.09	0.91
SPDW	cluster_3	Developed ex-US Equity	2026-04-10	31	6.1%	0.10	0.90
VEA	cluster_3	Developed ex-US Equity	2026-04-10	31	6.0%	0.11	0.89

IQDF	cluster_3	Developed ex-US Equity	2026-04-10	31	5.7%	0.16	0.84
IPKW	cluster_3	Developed ex-US Equity	2026-04-09	32	2.2%	0.21	0.79
FEP	cluster_3	Developed ex-US Equity	2026-04-09	32	3.9%	0.23	0.77
SCZ	cluster_3	Developed ex-US Equity	2026-04-10	31	4.3%	0.23	0.77
EWO	cluster_3	Developed ex-US Equity	2026-04-10	31	6.5%	0.23	0.77
CWI	cluster_3	Developed ex-US Equity	2026-04-10	31	5.8%	0.23	0.77
IMTM	cluster_3	Developed ex-US Equity	2026-04-10	31	3.5%	0.26	0.74
ACWX	cluster_3	Developed ex-US Equity	2026-04-10	31	5.9%	0.27	0.73
EWD	cluster_3	Developed ex-US Equity	2026-05-21	2	1.8%	0.21	0.71
VEU	cluster_3	Developed ex-US Equity	2026-04-10	31	6.2%	0.30	0.70
IXUS	cluster_3	Developed ex-US Equity	2026-04-10	31	6.1%	0.32	0.68
EWI	cluster_3	Developed ex-US Equity	2026-04-08	33	4.9%	0.36	0.64
VXUS	cluster_3	Developed ex-US Equity	2026-04-10	31	6.0%	0.36	0.64
DLS	cluster_3	Developed ex-US Equity	2026-04-13	30	1.5%	0.56	0.43
FNDC	cluster_3	Developed ex-US Equity	2026-04-10	31	3.6%	0.65	0.35
CIBR	cluster_16	Technology	2026-04-17	26	27.1%	0.00	1.00
HACK	cluster_16	Technology	2026-04-17	26	21.4%	0.00	1.00
IGV	cluster_16	Technology	2026-04-21	24	8.5%	0.00	1.00
IPO	cluster_16	Technology	2026-04-13	30	16.5%	0.06	0.94
SKYY	cluster_16	Technology	2026-04-17	26	14.0%	0.08	0.92
FDN	cluster_16	Technology	2026-04-10	31	12.7%	0.20	0.80
BLCN	cluster_16	Technology	2026-04-15	28	12.3%	0.37	0.63
BLOK	cluster_16	Technology	2026-04-13	30	16.8%	0.63	0.37
ARKW	cluster_16	Technology	2026-04-14	29	11.5%	0.69	0.31
LQDH	cluster_23	Bank Loan / Floating Ra...	2026-04-06	35	1.8%	0.00	1.00
IGBH	cluster_23	Bank Loan / Floating Ra...	2026-04-08	33	2.4%	0.00	1.00
IGHG	cluster_23	Bank Loan / Floating Ra...	2026-04-07	34	1.6%	0.00	1.00
SRLN	cluster_23	Bank Loan / Floating Ra...	2026-04-07	34	1.8%	0.95	0.05
USMV	cluster_2	U.S. Large Cap Value	2026-05-05	14	2.1%	0.00	1.00
SCHD	cluster_2	U.S. Large Cap Value	2026-04-21	24	5.5%	0.00	1.00
MGV	cluster_2	U.S. Large Cap Value	2026-04-13	30	5.5%	0.00	1.00
ACWV	cluster_2	U.S. Large Cap Value	2026-05-06	13	0.8%	0.00	1.00
HDV	cluster_2	U.S. Large Cap Value	2026-05-01	16	1.1%	0.00	1.00
DHS	cluster_2	U.S. Large Cap Value	2026-04-20	25	2.3%	0.00	1.00
ILCV	cluster_2	U.S. Large Cap Value	2026-04-13	30	5.8%	0.00	1.00
PWV	cluster_2	U.S. Large Cap Value	2026-04-02	36	6.2%	0.00	1.00
OUSA	cluster_2	U.S. Large Cap Value	2026-04-20	25	1.8%	0.00	1.00
DLN	cluster_2	U.S. Large Cap Value	2026-04-13	30	4.5%	0.00	1.00
VOOV	cluster_2	U.S. Large Cap Value	2026-04-14	29	4.8%	0.00	1.00
IVE	cluster_2	U.S. Large Cap Value	2026-04-14	29	4.7%	0.00	1.00
RDIV	cluster_2	U.S. Large Cap Value	2026-04-21	24	4.0%	0.00	1.00
SPYD	cluster_2	U.S. Large Cap Value	2026-04-21	24	3.1%	0.00	1.00
SPYV	cluster_2	U.S. Large Cap Value	2026-04-14	29	4.8%	0.00	1.00
DTD	cluster_2	U.S. Large Cap Value	2026-04-13	30	4.3%	0.00	1.00
VLUE	cluster_2	U.S. Large Cap Value	2026-04-09	32	27.3%	0.00	1.00

QDF	cluster_2	U.S. Large Cap Value	2026-04-13	30	7.8%	0.00	1.00
DIV	cluster_2	U.S. Large Cap Value	2026-04-07	34	2.8%	0.00	1.00
VTV	cluster_2	U.S. Large Cap Value	2026-04-13	30	4.7%	0.00	1.00
SDOG	cluster_2	U.S. Large Cap Value	2026-04-07	34	4.5%	0.00	1.00
DGRO	cluster_2	U.S. Large Cap Value	2026-04-13	30	3.2%	0.00	1.00
FTCS	cluster_2	U.S. Large Cap Value	2026-05-20	3	0.4%	0.01	0.99
FNDX	cluster_2	U.S. Large Cap Value	2026-04-09	32	7.4%	0.01	0.99
OMFL	cluster_2	U.S. Large Cap Value	2026-04-10	31	9.3%	0.01	0.99
FDVV	cluster_2	U.S. Large Cap Value	2026-04-14	29	4.9%	0.01	0.99
RWL	cluster_2	U.S. Large Cap Value	2026-04-10	31	6.5%	0.01	0.99
IUSV	cluster_2	U.S. Large Cap Value	2026-04-14	29	4.6%	0.01	0.99
IWX	cluster_2	U.S. Large Cap Value	2026-04-10	31	8.2%	0.01	0.99
SCHV	cluster_2	U.S. Large Cap Value	2026-04-10	31	6.7%	0.01	0.99
FDL	cluster_2	U.S. Large Cap Value	2026-05-19	4	-0.4%	0.01	0.99
PFM	cluster_2	U.S. Large Cap Value	2026-04-14	29	3.9%	0.02	0.98
RPV	cluster_2	U.S. Large Cap Value	2026-04-20	25	2.1%	0.02	0.98
VIG	cluster_2	U.S. Large Cap Value	2026-04-14	29	4.2%	0.02	0.98
SPHD	cluster_2	U.S. Large Cap Value	2026-05-18	5	1.1%	0.02	0.98
PRF	cluster_2	U.S. Large Cap Value	2026-04-09	32	8.2%	0.02	0.98
AIVL	cluster_2	U.S. Large Cap Value	2026-04-10	31	5.4%	0.08	0.92
IWD	cluster_2	U.S. Large Cap Value	2026-04-10	31	7.4%	0.09	0.91
VONV	cluster_2	U.S. Large Cap Value	2026-04-10	31	7.4%	0.09	0.91
VYM	cluster_2	U.S. Large Cap Value	2026-04-10	31	4.4%	0.10	0.90
NULV	cluster_2	U.S. Large Cap Value	2026-04-10	31	8.5%	0.10	0.90
DIA	cluster_2	U.S. Large Cap Value	2026-04-14	29	4.1%	0.11	0.89
JVAL	cluster_2	U.S. Large Cap Value	2026-04-10	31	11.9%	0.11	0.89
FVD	cluster_2	U.S. Large Cap Value	2026-05-22	1	-0.3%	0.08	0.88
DGRW	cluster_2	U.S. Large Cap Value	2026-04-14	29	4.7%	0.12	0.88
SDY	cluster_2	U.S. Large Cap Value	2026-05-20	3	1.4%	0.14	0.86
SPLV	cluster_2	U.S. Large Cap Value	2026-05-22	1	-0.3%	0.09	0.84
RSP	cluster_2	U.S. Large Cap Value	2026-04-14	29	4.0%	0.21	0.79
ONEY	cluster_2	U.S. Large Cap Value	2026-04-10	31	4.4%	0.21	0.79
LVHD	cluster_2	U.S. Large Cap Value	2026-05-22	1	-0.6%	0.15	0.77
FEX	cluster_2	U.S. Large Cap Value	2026-04-09	32	6.3%	0.24	0.76
SIZE	cluster_2	U.S. Large Cap Value	2026-04-13	30	5.5%	0.25	0.75
IMCB	cluster_2	U.S. Large Cap Value	2026-04-10	31	7.3%	0.35	0.65
FTA	cluster_2	U.S. Large Cap Value	2026-05-21	2	0.8%	0.36	0.58
IWR	cluster_2	U.S. Large Cap Value	2026-04-10	31	6.2%	0.54	0.46
NOBL	cluster_2	U.S. Large Cap Value	2026-05-22	1	-0.1%	0.52	0.41
VO	cluster_2	U.S. Large Cap Value	2026-04-13	30	4.8%	0.63	0.37
SMMV	cluster_2	U.S. Large Cap Value	2026-05-21	2	0.5%	0.66	0.26
PKW	cluster_2	U.S. Large Cap Value	2026-05-20	3	0.6%	0.95	-0.49
REGL	cluster_2	U.S. Large Cap Value	2026-05-26	0	0.0%	0.98	-0.60
FXR	cluster_2	U.S. Large Cap Value	2026-05-26	0	0.0%	0.99	-0.71
FIW	cluster_2	U.S. Large Cap Value	2026-05-08	11	-2.2%	1.00	-1.00

REZ	cluster_15	Real Estate	2026-04-14	29	4.7%	0.02	0.98
NURE	cluster_15	Real Estate	2026-04-13	30	7.5%	0.04	0.96
RWR	cluster_15	Real Estate	2026-04-10	31	5.7%	0.11	0.89
SCHH	cluster_15	Real Estate	2026-04-09	32	5.7%	0.12	0.88
PSR	cluster_15	Real Estate	2026-04-09	32	6.3%	0.14	0.86
USRT	cluster_15	Real Estate	2026-04-09	32	6.4%	0.15	0.85
ICF	cluster_15	Real Estate	2026-04-08	33	6.4%	0.23	0.77
REET	cluster_15	Real Estate	2026-04-13	30	4.1%	0.47	0.53
FV	cluster_6	U.S. Mid Cap Equity	2026-04-14	29	12.4%	0.00	1.00
XAR	cluster_6	U.S. Mid Cap Equity	2026-05-22	1	2.0%	0.05	0.90
ITA	cluster_6	U.S. Mid Cap Equity	2026-05-26	0	0.0%	0.03	0.87
MTUM	cluster_6	U.S. Mid Cap Equity	2026-04-08	33	21.0%	0.14	0.86
FTC	cluster_6	U.S. Mid Cap Equity	2026-04-10	31	11.8%	0.25	0.75
RPG	cluster_6	U.S. Mid Cap Equity	2026-04-09	32	16.1%	0.26	0.74
PTF	cluster_6	U.S. Mid Cap Equity	2026-04-07	34	39.8%	0.28	0.72
IWP	cluster_6	U.S. Mid Cap Equity	2026-04-15	28	4.1%	0.30	0.70
ICVT	cluster_6	U.S. Mid Cap Equity	2026-04-08	33	11.8%	0.30	0.70
IMCG	cluster_6	U.S. Mid Cap Equity	2026-04-09	32	11.4%	0.32	0.68
ARKQ	cluster_6	U.S. Mid Cap Equity	2026-04-14	29	13.1%	0.34	0.66
FPX	cluster_6	U.S. Mid Cap Equity	2026-04-08	33	14.3%	0.40	0.60
CWB	cluster_6	U.S. Mid Cap Equity	2026-04-08	33	10.9%	0.43	0.57
VBK	cluster_6	U.S. Mid Cap Equity	2026-04-10	31	11.7%	0.48	0.52
SCHM	cluster_6	U.S. Mid Cap Equity	2026-04-09	32	8.4%	0.57	0.43
IVOG	cluster_6	U.S. Mid Cap Equity	2026-04-09	32	8.1%	0.57	0.42
MDYG	cluster_6	U.S. Mid Cap Equity	2026-04-09	32	8.0%	0.60	0.40
IJK	cluster_6	U.S. Mid Cap Equity	2026-04-09	32	8.0%	0.61	0.39
XMMO	cluster_6	U.S. Mid Cap Equity	2026-04-06	35	14.8%	0.62	0.38
VOT	cluster_6	U.S. Mid Cap Equity	2026-04-14	29	6.4%	0.62	0.38
VFMO	cluster_6	U.S. Mid Cap Equity	2026-04-08	33	11.9%	0.62	0.38
VTWG	cluster_6	U.S. Mid Cap Equity	2026-04-13	30	12.2%	0.67	0.33
PDP	cluster_6	U.S. Mid Cap Equity	2026-04-08	33	10.9%	0.68	0.32
IWO	cluster_6	U.S. Mid Cap Equity	2026-04-13	30	12.0%	0.69	0.31
EWY	cluster_8	Emerging Markets Equity	2026-04-09	32	44.1%	0.12	0.88
AIA	cluster_8	Emerging Markets Equity	2026-04-10	31	23.2%	0.17	0.83
DEM	cluster_8	Emerging Markets Equity	2026-04-09	32	8.2%	0.28	0.72
EWT	cluster_8	Emerging Markets Equity	2026-04-08	33	33.8%	0.36	0.64
EEMV	cluster_8	Emerging Markets Equity	2026-04-10	31	10.2%	0.38	0.62
EMGF	cluster_8	Emerging Markets Equity	2026-04-10	31	14.2%	0.50	0.50
XSOE	cluster_8	Emerging Markets Equity	2026-04-10	31	14.1%	0.51	0.49
GEM	cluster_8	Emerging Markets Equity	2026-04-10	31	12.1%	0.51	0.49
AAXJ	cluster_8	Emerging Markets Equity	2026-04-13	30	14.5%	0.55	0.45
EEM	cluster_8	Emerging Markets Equity	2026-04-10	31	12.9%	0.61	0.39
ESGE	cluster_8	Emerging Markets Equity	2026-04-13	30	11.7%	0.61	0.39
IEMG	cluster_8	Emerging Markets Equity	2026-04-10	31	12.6%	0.66	0.34
EEMA	cluster_8	Emerging Markets Equity	2026-04-13	30	14.7%	0.70	0.30

SPEM	cluster_8	Emerging Markets Equity	2026-04-13	30	5.6%	0.95	0.05
USO	cluster_20	Broad Commodities	2026-01-29	80	73.1%	0.50	0.48
DBC	cluster_20	Broad Commodities	2025-12-22	105	34.1%	0.84	0.15
PDBC	cluster_20	Broad Commodities	2025-12-22	105	35.0%	0.84	0.15
KIE	cluster_14	Financials	2026-05-22	1	-0.6%	0.17	0.83
IAI	cluster_14	Financials	2026-04-10	31	5.4%	0.48	0.52
XLF	cluster_14	Financials	2026-04-13	30	0.4%	0.67	0.33
FNCL	cluster_14	Financials	2026-04-13	30	0.5%	0.76	0.24
KCE	cluster_14	Financials	2026-04-13	30	4.0%	0.98	0.02
IAT	cluster_14	Financials	2026-05-20	3	1.5%	0.81	-0.01
IYK	cluster_19	Consumer Staples	2026-05-06	13	0.4%	0.02	0.98
KXI	cluster_19	Consumer Staples	2026-05-06	13	-1.2%	0.02	0.98
XLP	cluster_19	Consumer Staples	2026-05-05	14	-0.5%	0.06	0.94
FSTA	cluster_19	Consumer Staples	2026-05-01	16	-2.2%	0.56	0.44
VDC	cluster_19	Consumer Staples	2026-05-01	16	-2.2%	0.57	0.43
FTXR	cluster_4	U.S. Small Cap Equity	2026-04-17	26	5.4%	0.11	0.89
ISMD	cluster_4	U.S. Small Cap Equity	2026-04-09	32	11.2%	0.24	0.76
SCHA	cluster_4	U.S. Small Cap Equity	2026-04-09	32	10.2%	0.61	0.39
IJS	cluster_4	U.S. Small Cap Equity	2026-04-10	31	6.6%	0.85	0.15
PEJ	cluster_4	U.S. Small Cap Equity	2026-05-26	0	0.0%	0.57	-0.08
FXD	cluster_4	U.S. Small Cap Equity	2026-05-06	13	-0.4%	0.74	-0.65
XHB	cluster_4	U.S. Small Cap Equity	2026-05-06	13	-3.2%	0.99	-0.98
NORW	cluster_9	Materials	2025-12-09	114	34.4%	0.00	1.00
MLPX	cluster_9	Materials	2026-04-30	17	-0.4%	0.01	0.99
AMLX	cluster_9	Materials	2026-04-29	18	0.4%	0.04	0.96
MLPA	cluster_9	Materials	2026-04-29	18	0.3%	0.05	0.95
EMLP	cluster_9	Materials	2026-05-14	7	0.1%	0.07	0.91
DBB	cluster_9	Materials	2026-04-10	31	7.3%	0.18	0.82
IGE	cluster_9	Materials	2026-05-14	7	-1.5%	0.26	0.48
SLX	cluster_9	Materials	2026-04-09	32	12.1%	0.67	0.33
COPX	cluster_9	Materials	2026-05-08	11	1.8%	0.53	0.30
IYM	cluster_9	Materials	2026-05-26	0	0.0%	0.90	-0.42
XME	cluster_9	Materials	2026-05-26	0	0.0%	0.97	-0.44
HYEM	cluster_10	High Yield Bond	2026-04-13	30	1.0%	0.94	0.06
VWO	cluster_10	High Yield Bond	2026-05-26	0	0.0%	1.00	-0.61
CNXT	cluster_12	China Equity	2026-04-10	31	18.2%	0.28	0.72
CQQQ	cluster_12	China Equity	2026-04-20	25	7.6%	0.30	0.70
PID	cluster_7	Global Equity	2026-04-30	17	1.5%	0.00	1.00
EWL	cluster_7	Global Equity	2026-04-15	28	2.0%	0.01	0.99
DWX	cluster_7	Global Equity	2026-05-04	15	1.6%	0.02	0.98
EFAV	cluster_7	Global Equity	2026-05-20	3	-1.2%	0.02	0.89
NFRA	cluster_7	Global Equity	2026-04-09	32	2.2%	0.13	0.87
EWG	cluster_7	Global Equity	2026-04-09	32	4.0%	0.34	0.66
EWI	cluster_7	Global Equity	2026-05-05	14	1.7%	0.52	0.35
IGF	cluster_7	Global Equity	2026-05-26	0	0.0%	0.44	0.19

ENZL	cluster_7	Global Equity	2026-05-22	1	0.8%	0.82	-0.05
BBAX	cluster_7	Global Equity	2026-04-30	17	-0.9%	0.81	-0.12
ECH	cluster_7	Global Equity	2026-05-06	13	-1.4%	0.83	-0.78
FXB	cluster_25	International Fixed Inc...	2026-05-22	1	0.1%	0.99	-0.60
HYD	cluster_17	Municipal Bond	2026-05-22	1	0.5%	0.99	-0.49
HYMB	cluster_17	Municipal Bond	2026-05-26	0	0.0%	0.99	-0.69
SUB	cluster_17	Municipal Bond	2026-05-11	10	0.0%	0.97	-0.91
IAGG	cluster_5	Aggregate Fixed Income	2026-05-22	1	0.3%	0.31	0.56
BNDX	cluster_5	Aggregate Fixed Income	2026-05-26	0	0.0%	0.37	0.43
IGSB	cluster_5	Aggregate Fixed Income	2026-05-26	0	0.0%	0.95	-0.50
VCSH	cluster_5	Aggregate Fixed Income	2026-05-26	0	0.0%	0.97	-0.65
AGZ	cluster_5	Aggregate Fixed Income	2026-05-11	10	-0.1%	1.00	-1.00

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
HACK	cluster_16	Technology	1.00	1.00
MINT	cluster_24	Cash / Ultra-Short Fixed ...	1.00	1.00
FLOT	cluster_23	Bank Loan / Floating Rate	1.00	1.00
IBDR	cluster_5	Aggregate Fixed Income	1.00	1.00
USMV	cluster_2	U.S. Large Cap Value	1.00	1.00
QQEW	cluster_1	U.S. Large Cap Equity	1.00	1.00
FV	cluster_6	U.S. Mid Cap Equity	1.00	1.00
XLV	cluster_11	Healthcare	1.00	1.00
EWJV	cluster_22	Japan Equity	1.00	1.00
PID	cluster_7	Global Equity	1.00	1.00
NORW	cluster_9	Materials	1.00	1.00
DBEF	cluster_3	Developed ex-US Equity	1.00	1.00
IYK	cluster_19	Consumer Staples	1.00	0.98
REZ	cluster_15	Real Estate	1.00	0.98
FTXR	cluster_4	U.S. Small Cap Equity	1.00	0.89
EWY	cluster_8	Emerging Markets Equity	1.00	0.88
KIE	cluster_14	Financials	0.99	0.83
CNXT	cluster_12	China Equity	1.00	0.72
DBO	cluster_20	Broad Commodities	1.00	0.71
HYS	cluster_10	High Yield Bond	0.97	0.21

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
EIDO	cluster_7	Global Equity	1.00	-1.00
PPLT	cluster_26	Currency	1.00	-1.00
CHIQ	cluster_12	China Equity	1.00	-1.00
PGF	cluster_10	High Yield Bond	1.00	-1.00
REM	cluster_2	U.S. Large Cap Value	1.00	-1.00
BAB	cluster_13	Fixed Income	1.00	-1.00

PZA	cluster_17	Municipal Bond	1.00	-1.00
SPTI	cluster_5	Aggregate Fixed Income	1.00	-1.00
URA	cluster_9	Materials	1.00	-1.00
PBJ	cluster_19	Consumer Staples	0.99	-0.99
PKB	cluster_4	U.S. Small Cap Equity	1.00	-0.99
ESPO	cluster_1	U.S. Large Cap Equity	0.99	-0.99
LTPZ	cluster_21	Inflation-Protected Fixed...	0.98	-0.97
DVYE	cluster_8	Emerging Markets Equity	0.99	-0.97
IAU	cluster_18	Gold	0.97	-0.96
FXE	cluster_25	International Fixed Income	1.00	-0.95
IBUY	cluster_16	Technology	0.96	-0.94
BKLN	cluster_23	Bank Loan / Floating Rate	1.00	-0.84
GSY	cluster_24	Cash / Ultra-Short Fixed ...	1.00	-0.74
IBB	cluster_11	Healthcare	0.79	-0.61

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_26	5	FLAT	0.302	0.094	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_26	5	FLAT	0.493	0.076	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.689	0.042	0.00007	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.732	0.096	0.00022	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.614	0.222	0.00035	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.765	0.137	0.00031	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_20	10	FLAT	0.596	0.171	0.00040	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.822	0.023	0.00001	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.822	0.092	0.00022	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.642	0.190	0.00047	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.767	0.169	0.00039	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.720	0.189	0.00043	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_26	5	FLAT	0.492	0.216	0.00055	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.767	0.180	0.00043	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	43	FLAT	0.844	0.128	0.00034	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.818	0.203	0.00048	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_5	43	FLAT	0.877	0.162	0.00030	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_5	43	FLAT	0.836	0.235	0.00046	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_17	13	FLAT	0.878	0.250	0.00045	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_7	30	FLAT	0.447	0.331	0.00083	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_17	13	LONG	0.878	0.265	0.00052	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_12	15	FLAT	0.806	0.270	0.00071	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_9	21	LONG	0.751	0.358	0.00083	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_7	30	FLAT	0.501	0.367	0.00105	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_12	15	FLAT	0.855	0.255	0.00071	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_7	30	FLAT	0.579	0.412	0.00107	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.865	0.332	0.00071	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	FLAT	0.919	0.226	0.00056	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_12	15	LONG	0.905	0.240	0.00063	
IHF	iShares U.S. Health Care Provid...	Healthcare	cluster_11	17	LONG	0.737	0.424	0.00113	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20